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Benchmarking Error Mitigation Strategies for the Variational Quantum Eigensolver on the One-Dimensional Transverse-Field Ising Model

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Abstract

Quantum error mitigation is essential for extracting accurate results from noisy intermediate-scale quantum (NISQ) hardware, but the relative effectiveness of different techniques remains poorly characterised. This work benchmarks all eight combinations of readout correction, parity post-selection, and zero-noise extrapolation (ZNE) for the variational quantum eigensolver (VQE) applied to the one-dimensional transverse-field Ising model, across three quantum phases, system sizes up to 12 qubits, and three ansatz depths, using noise models calibrated from the IBM Eagle r3 and Heron r2 processors. No universal optimal strategy is found: the best configuration depends jointly on the symmetry structure of the target Hamiltonian's ground state and the noise composition of the device. Readout correction combined with parity post-selection is optimal in the ordered phase, while readout correction combined with ZNE is optimal in the disordered phase. A parametric noise study further reveals that ZNE effectiveness follows a sigmoidal dependence on the device's gate-to-readout error ratio, with a crossover near $r \approx 0.4$ – 0.5 . This relationship constitutes a predictive tool: it explained the null ZNE result on the readout-dominated Eagle r3 processor and predicted a moderate improvement on Heron r2 before that benchmark was run. Both the predicted improvement and the broader simulation conclusions were confirmed by experiments on the `ibm_fez` quantum processor.

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1 Introduction

Quantum computers offer transformative capabilities by exploiting quantum phenomena, such as quantum superposition and entanglement, to solve problems intractable for classical machines. However, current quantum devices operate in the noisy intermediate-scale quantum (NISQ) regime [1, 2]. These processors are characterised by qubit counts of $O(10^2)$ and per-gate error rates of $10^{-3} - 10^{-2}$. These hardware limitations mean that the output of any quantum computation is corrupted by noise, limiting the practical utility of NISQ devices.

The variational quantum eigensolver (VQE) is considered the leading algorithm for ground state energy estimation on NISQ hardware [3]. It is a hybrid quantum-classical algorithm that uses shallow parametrised circuits on the quantum processor and delegates the iterative parameter optimisation to a classical computer. The shallow circuit depth also makes VQE circuits amenable to error mitigation techniques, which are most effective when the total accumulated noise remains moderate. VQE was originally proposed for molecular systems and has since been applied to condensed matter and spin models [4, 5].

The target system in this study is the 1D transverse-field Ising model (TFIM). This model is a paradigmatic quantum many-body system exhibiting a quantum phase transition at $h/J = 1$ between ordered and disordered phases [6]. Crucially, it is exactly solvable for finite chains via exact diagonalisation, providing an unambiguous reference energy against which VQE results can be compared [7]. This makes the TFIM an ideal benchmark for evaluating the accuracy of quantum algorithms and the effectiveness of error mitigation strategies.

On NISQ hardware, noise pushes VQE measured energies away from the true ground state, degrading accuracy. There are two approaches to address hardware noise: quantum error correction (QEC), and quantum error mitigation (QEM) [8, 9]. QEC encodes logical qubits into many physical qubits with additional measurement overhead, requiring qubit counts that exceed current device capabilities. QEM instead uses classical post-processing and controlled noise amplification to reduce the impact of errors without additional qubits. This study benchmarks three error mitigation techniques: readout correction, which addresses measurement noise [5]; parity post-selection, which exploits the $\mathbb{Z}_2$ symmetry of the TFIM Hamiltonian [10]; and zero-noise extrapolation (ZNE), which targets gate errors via controlled noise amplification and polynomial extrapolation [11].

This study systematically benchmarks all $2^3 = 8$ subsets of three mitigation methods across 54 (N, h, L) parameter configurations, using two IBM noise models (FakeBrisbane/Eagle r3 and FakeFez/Heron r2), with real hardware validation on `ibm_fez`. The exhaustive combinatorial approach allows isolating both individual and synergistic contributions of each method. The phase-dependent Pareto analysis in Section 3 reveals that no single strategy is universally optimal,

and the parametric noise composition study in Section 4 explains the ZNE null result on Brisbane and predicts its effectiveness on Fez before running the benchmark. However, the study does not explore full QEC, system sizes beyond $N = 12$, or alternative variational algorithms.

The contributions of this work are threefold. First, an exhaustive benchmark of all eight subsets of three mitigation methods across 54 (N, h, L) configurations and two IBM noise models, demonstrating that the optimal strategy is phase dependent. Second, a parametric study revealing a sigmoidal dependence of zero-noise extrapolation effectiveness on the gate-to-readout error ratio, with crossover near $r \approx 0.4$ – 0.5 . Third, real hardware validation on `ibm.fez` confirming the simulation predictions. While Sobrino et al. [12] independently identify qubit error probability as relevant to ZNE in a complementary framework, the present work embeds this finding within a full comparative mitigation benchmark across quantum phases and uses the noise-ratio framework to predict ZNE behaviour across two distinct device architectures before benchmarking.

The remainder of this paper is organised as follows. Section 2 presents the theoretical background. Section 3 presents the main benchmark results on FakeBrisbane. Section 4 presents the ZNE noise composition study. Section 5 presents the FakeFez comparison and hardware validation. Section 6 summarises the findings and discusses future work.

2 Theory

2.1 The 1D Transverse-Field Ising Model

The Transverse-Field Ising Model (TFIM) is a canonical model of quantum many-body physics because it captures the essential behaviour of quantum criticality in the simplest possible setting [6, 7]. The TFIM is defined as a chain of N spin-1/2 particles with nearest-neighbour interactions and a transverse magnetic field. Subject to open boundary conditions (OBC), the $\sigma_i^z \sigma_{i+1}^z$ interaction couples sites i and $i + 1$ for $i = 1, \dots, N - 1$ with no periodic wrapping, which is the standard choice for studies of small N [7]. The Hamiltonian is given by:

$$H = -J \sum_{i=1}^{N-1} \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^N \sigma_i^x. \quad (1)$$

The Hamiltonian consists of two competing terms. The first term represents the nearest-neighbour ferromagnetic spin coupling, where $J > 0$ is the exchange coupling constant, which energetically favours spin alignment along the z -direction. The second term is the transverse magnetic field, where $h \geq 0$ is the transverse field strength, which induces quantum tunnelling from the σ^z eigenstates, driving the system away from classical order towards a quantum paramagnetic state. Since operators

σ^x and σ^z are non-commuting, there is no basis in which both terms in the Hamiltonian are simultaneously diagonal, thus the ground state is a non-trivial quantum superposition that cannot be expressed as a product state, except in the limiting cases. The competition of these non-commuting terms characterises the ground state, depending entirely on the ratio h/J . This gives rise to three distinct phases. In the limit $J \gg h$, the system is in the ordered ferromagnetic phase, where the spin interaction dominates. In this phase, for finite N and any $h > 0$, the ground state is the symmetric superposition $\frac{1}{\sqrt{2}}(|\uparrow\uparrow\cdots\uparrow\rangle + |\downarrow\downarrow\cdots\downarrow\rangle)$, a Greenberger-Horne-Zeilinger (GHZ) state, with corrections that vanish as $h/J \rightarrow 0$ [6, 7, 13]. This state exhibits perfect long-range spin correlations, $\langle\sigma_i^z\sigma_j^z\rangle = 1$ for all i, j , and carries exclusively even $\mathbb{Z}_2$ parity. In the limit $h \gg J$, the transverse field energetically dominates, and the ground state approaches $|+\rangle^{\otimes N}$, where each spin is the positive eigenstate of σ^x [6, 7]. This is the disordered paramagnetic phase. At the quantum critical point $h/J = 1$ [7], neither term dominates and the system is at the boundary between phases, where quantum fluctuations are maximal [6, 7]. For finite N , the sharp phase transition is replaced by a smooth crossover, with the critical features becoming more pronounced as N increases [6]. The phase structure is summarised schematically in Figure 1. The TFIM Hamiltonian additionally conserves the global $\mathbb{Z}_2$ parity, defined by the operator $P = \bigotimes_{i=1}^N \sigma_i^x$, which commutes with H and is exploited as an error mitigation strategy in Section 2.5.2 [10].

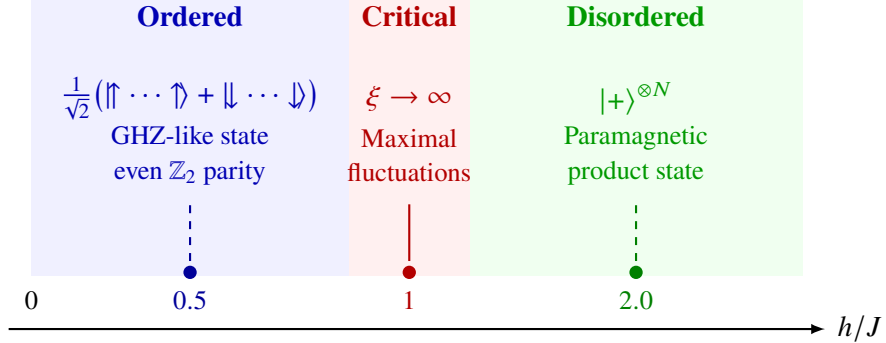


Figure 1: Schematic phase diagram of the 1D transverse-field Ising model. The ordered phase ($h/J < 1$) has a GHZ-like ground state with even $\mathbb{Z}_2$ parity and perfect spin correlations. At the quantum critical point $h/J = 1$, the correlation length ξ diverges. In the disordered phase ($h/J > 1$), the ground state approaches the paramagnetic product state $|+\rangle^{\otimes N}$. The three values of h/J studied in this work (filled circles) sample each phase.

The three values of h/J studied in this work, $\{0.5, 1.0, 2.0\}$, were deliberately chosen to examine the ordered, critical, and disordered phases respectively, to test whether mitigation performance is sensitive to quantum phase. Throughout this paper, $J = 1$ is fixed, reducing the ratio h/J to simply h .

2.2 Exact Diagonalisation

Although the 1D TFIM is analytically solvable via the Jordan-Wigner transformation [7, 14], this work uses numerical exact diagonalisation (ED) because it generalises to other Hamiltonians and provides the ground state energy to floating-point precision for the finite chains studied.

The N -site TFIM lives in a Hilbert space of dimension 2^N , spanned by the 2^N computational basis states $|s_1 s_2 \cdots s_N\rangle$ where $s_i \in \{0, 1\}$ represents spin up or spin down at site i . Additionally, each Pauli operator σ^z, σ^x has a 2×2 matrix representation, and the full N -site operators σ_i^z, σ_i^x are constructed as tensor products of these with identity operators I on all other sites, explicitly $\sigma_i^z = I^{\otimes(i-1)} \otimes \sigma^z \otimes I^{\otimes(N-i)}$. The full Hamiltonian matrix is therefore a $2^N \times 2^N$ Hermitian operator, constructed by summing these tensor product terms according to Equation (1). Since the matrix is Hermitian, its eigenvalues are real, and the ground state energy, E_0 , is its minimum eigenvalue. In this study, Qiskit’s `SparsePauliOp` class was used to construct the Hamiltonian matrix, as it represents the Hamiltonian efficiently as a sum of Pauli strings, avoiding the $O(4^N)$ memory cost of explicit dense construction until diagonalisation is required [15].

To obtain the ground state energy, the $2^N \times 2^N$ matrix is diagonalized using a standard numerical eigensolver. Since only the smallest eigenvalue is required, the computation uses `numpy.linalg.eigvalsh`, which exploits Hermitian symmetry and omits the eigenvector computation, making it faster and more memory-efficient than a full eigendecomposition [16]. Since the Hamiltonian matrix is constructed exactly from the Pauli operator definitions and diagonalised numerically without further approximation, the minimum eigenvalue obtained constitutes the exact ground state energy E_0 of the finite chain. Unlike variational methods, which produce an upper bound on E_0 , ED introduces no systematic error, making it the unambiguous reference against which all VQE results in this work are measured.

The computational cost of ED scales exponentially with system size [3, 17]. The Hamiltonian matrix has $2^N \times 2^N$ elements, requiring $O(4^N)$ memory for dense storage. Diagonalisation via standard Hermitian eigensolver routines scales as $O(8^N) = O((2^N)^3)$ in time. For example, for $N = 10$, the matrix has dimension 1024×1024 and is entirely tractable on a standard computer. For $N = 50$ the matrix has dimension $2^{50} \approx 10^{15}$, requiring memory on the order of petabytes, far beyond any existing classical hardware. This exponential wall renders ED infeasible beyond $N \sim 20 - 30$ on classical hardware, yet it is precisely tractable for the small system sizes accessible to current NISQ devices, making it an ideal classical benchmark for this regime. The exponential cost of ED for large N therefore provides the fundamental motivation for quantum simulation algorithms such as VQE, introduced in Section 2.3 [3].

2.3 Variational Quantum Eigensolver

The Variational Quantum Eigensolver (VQE) is a hybrid quantum-classical algorithm first proposed by Peruzzo et al. to estimate molecular ground state energies [4]. VQE has since been extended beyond molecular problems to condensed matter and spin models [5, 18]. The VQE operates by preparing and measuring parametrized quantum states on a quantum processor and optimising the parameters using a classical iterative routine until the measured energy reaches a minimum. This task division ensures that the circuit stays shallow, while the repetitive and computationally cheap task is performed classically. Because the quantum processor operates on low-depth circuits, which accumulate less noise as will be discussed in Section 2.4, the VQE has become the standard NISQ-era algorithm for ground state problems [3]. The VQE is the application of the Rayleigh-Ritz variational method to quantum computing, which is introduced in Section 2.3.1. The hybrid loop pipeline is illustrated in Figure 2; each component is defined formally in the following subsections.

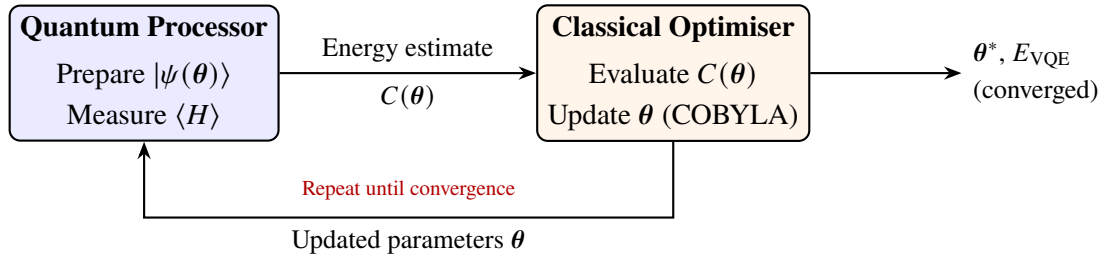


Figure 2: Schematic of the VQE hybrid quantum-classical loop. The quantum processor prepares the parametrised state $|\psi(\theta)\rangle$ and measures $\langle H \rangle$, returning the energy estimate $C(\theta)$. The classical optimiser (COBYLA) updates θ to minimise $C(\theta)$. The loop terminates at convergence, yielding the optimised parameters θ^* and the VQE ground state energy estimate.

2.3.1 The Variational Principle

For any normalized trial state in a quantum circuit $|\psi\rangle$, the expectation value of the Hamiltonian will be an upper bound on the true ground state energy E_0 [3, 19, 20]:

$$\langle \psi | H | \psi \rangle \geq E_0. \quad (2)$$

The expectation value is only equal to E_0 if $|\psi\rangle$ is the ground state. Hence, the expectation value of H with respect to any parametrised state can only overestimate or equal the ground state energy. Optimising the parameters systematically to minimise the expectation value can only approach E_0 from above.

In VQE, a parametrized quantum circuit $U(\theta)$ acts on the initial state $|0\rangle^{\otimes N}$, resulting in $|\psi(\theta)\rangle = U(\theta) |0\rangle^{\otimes N}$ [19]. The expectation value of the Hamiltonian is the cost function:

$$C(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle . \quad (3)$$

VQE minimises the cost function finding the parameters that set the circuit closest to the ground state, $\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \{C(\boldsymbol{\theta})\}$. Therefore, $C(\boldsymbol{\theta}^*)$ is the VQE estimate of E_0 . The variational principle guarantees that $C(\boldsymbol{\theta}^*) \geq E_0$.

As shown in Section 2.2, the Hamiltonian is a sum of Pauli strings, each being a tensor product of σ^x , σ^z , and I operators. Expectation values of $\sigma_i^z \sigma_{i+1}^z$ can be measured directly in the computational basis. However, measuring expectation values of σ^x requires a basis rotation. A Hadamard gate is applied before measurement to rotate from the X -eigenbasis to the Z -eigenbasis [19]. Thus, two separate circuits are needed per evaluation, one per eigenbasis [3]. The expectation value of the Hamiltonian is split into two components:

$$\langle H \rangle = \langle H_{ZZ} \rangle + \langle H_X \rangle . \quad (4)$$

Each circuit is measured a finite number of times introducing shot noise. The standard deviation of the sample mean of an observable measured in N_{shots} shots scales as $\sigma \propto 1/\sqrt{N_{\text{shots}}}$ [21].

2.3.2 Hardware-Efficient Ansatz

The design of the circuit constructed in the quantum processor, known as the ansatz, has a non-negligible effect on how algorithms acting on it perform. Broadly, there are two common ansatz approaches: problem-inspired ansatz, and hardware-efficient ansatz [3, 22]. A problem-inspired ansatz encodes physical knowledge of the target state, proving more precise, but requires complex circuits. Hardware-efficient ansatz (HEA) only uses gates native to the device used, regardless of the problem. This study uses HEA because it is easier to scale in size, provides a more general-purpose benchmark, and, since it is hardware native, compilation overhead is minimised [5].

Each circuit consists of L repeated layers, where each layer l consists of two blocks: (i) a rotation block of single-qubit $R_y(\theta)$ gates applied to every qubit, and (ii) an entangling block of CZ gates, which are native to the IBM heavy-hex architecture [15], applied to all adjacent qubit pairs $(q_0 - q_1, q_1 - q_2, \dots, q_{N-2} - q_{N-1})$, as illustrated in Figure 3 and defined in Equation (5) [5]. Layer U_1 is applied first (leftmost in the circuit) so that $|\psi(\boldsymbol{\theta})\rangle = U_L \dots U_1 |0\rangle^{\otimes N}$. The total number of variational parameters per circuit is then $N \times L$:

$$U(\boldsymbol{\theta}) = U_L \cdots U_2 U_1, \quad U_l = \text{CZ}_{\text{chain}} \left[\bigotimes_{q=0}^{N-1} R_y(\theta_{l,q}) \right]. \quad (5)$$

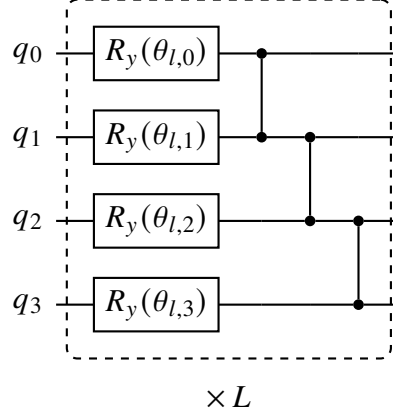


Figure 3: Hardware-efficient ansatz for $N = 4$ qubits. Each layer l consists of $R_y(\theta_{l,q})$ single-qubit rotations followed by a linear chain of CZ entangling gates between adjacent qubits. The dashed box is repeated L times with independent parameters per layer, giving $N \times L$ total variational parameters.

Many HEA use both a R_y and R_z rotation gate per layer, giving full coverage of the Bloch sphere to each qubit. Since both σ^x and σ^z are real matrices, the TFIM Hamiltonian has only purely real matrix elements. Hence, the trial states in the circuit can be fixed to only have purely real amplitudes. This permits omitting the R_z rotation, which adds complex phases. This restriction reduces the parameter count and circuit complexity without affecting expressibility for this particular model [21]. For completeness, the matrix representations of the $R_y(\theta)$ and CZ gates are

$$R_y(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}, \quad (6)$$

and

$$\text{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (7)$$

The expressibility of a particular ansatz is the fraction of the Hilbert space it can reach as $\boldsymbol{\theta}$ is varied [3]. With ansatz depth $L = 1$, correlations only reach nearest-neighbour distance. At depth L ,

correlations can propagate up to L sites along the chain. Hence, increasing ansatz depth improves expressibility. Furthermore, for the TFIM ground state, the correlation length diverges at criticality $h/J = 1$, so more expressibility is required near the critical point [6]. However, increasing depth requires adding gates, which increase noise on real hardware [3]. This relation between ansatz depth and noise is precisely one that this study covers across $L \in \{1, 2, 3\}$. For $N = 2$, even $L = 1$ provides sufficient expressibility to exactly represent the ground state, making $N = 4$ the smallest non-trivial benchmarking configuration.

2.3.3 Classical Optimisation via COBYLA

The cost function $C(\theta)$ evaluated in a noisy simulator is stochastic due to finite shot noise and gate noise. Gradient-based optimisers estimate $\partial C / \partial \theta_k$ for each of the $N \times L$ parameters θ_k . On noisy backends, the gradient estimates are dominated by noise rather than the true landscape, causing gradient-based methods to stall [21]. On quantum hardware, gradients can be obtained analytically via the parameter-shift rule, which uses two evaluations per parameter per gradient step, that is $2NL$ circuit evaluations per gradient step [23]. However, each circuit evaluation returns a noisy estimate of $C(\theta)$, so even analytically exact gradients produce noise-corrupted estimates. Derivative-free optimisers avoid this entirely by using function values to navigate the landscape, thus they are the preferred approach.

The Constrained Optimization By Linear Approximation (COBYLA), developed by Powell in 1994 [24], and implemented via SciPy’s `minimize` [25], constructs a linear approximation of the objective function within a trust region, of initial radius ρ_0 in parameter space. At each iteration, $C(\theta)$ is evaluated at a set of interpolation points, the function values are fitted to a local linear model, and the parameters are moved towards lower values within the trust region, which shrinks as the algorithm converges. COBYLA does not compute or estimate any derivatives. For this study, the initial trust region radius is set to $\rho_0 = \pi/4$, the standard for angular parameters spanning $[-\pi, \pi]$, and the maximum number of function evaluations is limited to `maxiter` = 600 per optimisation.

The COBYLA method finds local minima, not necessarily global minima. This motivates the decision to run the optimisation $n_{\text{restarts}} = 3$ times with independent random initial parameters θ_0 drawn uniformly from $[-\pi, \pi]^{NL}$. The restart yielding the lowest energy is retained. The multiple restarts strategy also guards against regions of exponentially vanishing gradients, known as barren plateaus, that can in principle occur in parametrised quantum circuits [3, 26, 27]. For the parameter counts in this study (at most $N \times L = 36$ variational parameters), barren plateaus are not expected to be a significant obstacle, as the onset of exponentially vanishing gradients typically requires much deeper circuits or global cost functions [26]. During the COBYLA optimisation process, a reduced shot count of $N_{\text{shots}}^{\text{opt}} = 1024$ is used, compared to $N_{\text{shots}}^{\text{eval}} = 8192$ for the final energy

evaluation. The optimisation needs to rank parameter vectors, not compute the precise cost function, so lower-precision evaluations suffice. Once the optimised parameters θ^* are found, the final reported energy is re-evaluated with $N_{\text{shots}}^{\text{eval}}$ and $n_{\text{reps}} = 10$ independent repetitions for statistical analysis, as described in Appendix A.

2.4 Noise in NISQ Devices

When performed on an ideal quantum computer, the VQE algorithm would return $C(\theta^*) = E_0$ exactly, assuming the ansatz is sufficiently expressive. On real quantum hardware, however, every gate operation, idle period, and measurement introduces errors that corrupt the quantum state. This pushes the measured energy away from the true theoretical value. Current quantum devices operate in the noisy intermediate-scale quantum (NISQ) regime [1]. NISQ devices are characterised by qubit counts of the order of $10^1 - 10^3$ and gate error rates of the order $10^{-3} - 10^{-2}$, without the circuit overhead of error correction. In order to design techniques that mitigate these errors, the source and nature of the hardware’s noise must be understood and analysed.

Two of the principal noise sources are relaxation time, T_1 , and dephasing time, T_2 . The quantum state of a prepared qubit degrades over time even if no operations are applied. T_1 characterises the timescale on which an excited qubit spontaneously decays from $|1\rangle$ to $|0\rangle$, releasing its energy to the environment. T_2 characterises the timescale on which quantum superposition coherence is lost and the relative phase between $|0\rangle$ and $|1\rangle$ components randomises. The relation between T_1 and T_2 is given by $T_2 \leq 2T_1$ [3, 13]. For IBM’s Brisbane quantum processor, which was used as the noise model for this study, typical values are $T_1 \sim 200 - 300 \mu\text{s}$ and $T_2 \sim 100 - 200 \mu\text{s}$ [15]. Gate operations take $\sim 30 - 60 \text{ ns}$ for single-qubit gates and $\sim 300 - 600 \text{ ns}$ for two-qubit gates [15, 28]. Hence, circuits of depth $\sim 100 - 1000$ can execute before decoherence dominates. Deeper circuits, that is circuits with larger L , accumulate more idle time, making decoherence noise more severe. This is consistent with the depth–noise trade-off discussed in Section 2.3.2. For the ansatz depths studied in this work ($L \leq 3$), the total transpiled circuit depth remains well within the coherence limit, thus decoherence is expected to be a secondary noise source relative to gate and readout errors.

Additionally, every gate operation also introduces error. In simulators, these errors are modeled as noise channels applied after each ideal gate operation. For single-qubit gates (RZ, SX, and X in IBM’s native gate set), typical error rates are $\sim 10^{-4} - 10^{-3}$. For two-qubit gates (CX), error rates are significantly higher, typically $\sim 10^{-3} - 10^{-2}$ [5]. On IBM’s recent heavy-hex architecture, where each qubit connects to at most three neighbours, two-qubit gate error rates are comparable to or smaller than readout error rates ($\sim 1 - 3\%$), meaning that for shallow circuits readout noise may dominate the noise profile [15]. Since the TFIM HEA requires $N - 1$ CZ gates per layer, and these are the dominant gate noise source, two-qubit gate errors accumulate approximately linearly

with circuit depth L and system size N . In the FakeBrisbane noise model used in this work, gate errors are modeled as depolarizing channels, which with probability p replace the output state with the maximally mixed state I/d , which represents a state of complete uncertainty with no preferred measurement outcome;

$$\mathcal{E}(\rho) = (1 - p)\rho + \frac{p}{d}I, \quad (8)$$

where ρ is the input density matrix of the qubits, p is the depolarising probability and d is the gate's dimension ($d = 2$ for single-qubit and $d = 4$ for two-qubit). For a traceless Hamiltonian, the maximally mixed state I/d has $\langle H \rangle = \text{Tr}(H)/2^N \approx 0$ [13]. Therefore, depolarising noise biases the measured energy systematically towards zero, away from the negative ground state energy.

The final source of error occurs during measurement events. When a qubit is prepared in state $|0\rangle$, there is a probability $p_{0 \rightarrow 1}$ of reading 1, and vice versa. Typical readout error rates on IBM hardware are $\sim 1 - 3\%$ per qubit [5]. For an N -qubit measurement, readout errors compound. A bitstring measurement outcome can have a subset of its bits flipped. Since each qubit has independent $p_{1 \rightarrow 0}$ and $p_{0 \rightarrow 1}$, readout errors introduce a systematic bias in the measurement statistics from which $\langle H \rangle$ is obtained [5, 29]. Unlike gate errors, readout errors occur after the quantum computation: they do not corrupt the state of the qubits, and are therefore amenable to classical post-processing correction. This is the basis of the readout error mitigation method, shown in Section 2.5.1.

The main benchmark in this study uses Qiskit's FakeBrisbane backend, which provides a noise model calibrated from IBM's 127-qubit Brisbane processor [15]. The noise model is extracted automatically from the FakeBrisbane backend via Qiskit's `NoiseModel.from_backend()`, which captures a snapshot of error rates calibrated from real hardware. This model includes T_1/T_2 decoherence noise, per-gate depolarising errors, and per-qubit readout errors, all extracted from real device calibration data. This allows for a realistic simulation environment without additional complications of queue times, dynamic noise profiles, and non-reproducibility inherent to real-hardware runs. However, the real noise profile on quantum hardware is dynamic, meaning that the noise models can only represent an instantaneous noise profile. The three noise sources described above degrade VQE accuracy through distinct mechanisms, motivating the error mitigation strategies presented in Section 2.5. Furthermore, each mitigation method can be engineered to target a specific noise channel. Readout errors are addressed by calibration-based correction, gate errors and decoherence by Zero-Noise Extrapolation, and errors that violate known symmetries of the Hamiltonian by parity post-selection, each described in Section 2.5.1, Section 2.5.3, and Section 2.5.2, respectively.

In this work a comparative study between the Brisbane processor (Eagle r3) and the Fez processor (Heron r2) is also carried out. The Heron processor is IBM's newest architecture, and it has a different

noise profile to the Eagle processors, as described in Table 1. Each noise profile is characterised by its noise ratio, defined as $r = p_{\text{gate}}/p_{\text{readout}}$. If the ratio is close to 1, the noise profile is balanced, if it is biased towards zero, it is readout-dominated, and vice versa. These differences will be discussed in Section 5.

	Eagle r3	Heron r2
Qubit count	127	156
Readout error	2.51%	0.76%
Gate Error	0.78%	0.39%
Noise ratio	0.31	0.51

Table 1: Noise profile comparison between IBM’s Eagle r3 and Heron r2 processors

2.5 Error Mitigation Methods

This section describes the three error mitigation methods benchmarked in this work. Each targets a different noise mechanism identified in Section 2.4, and none require the qubit overhead of full quantum error correction [9].

2.5.1 Readout Error Correction

For a single qubit q , readout noise is characterised by a 2×2 assignment matrix A_q whose elements are conditional probabilities: $A_q[i, j] = P(\text{measure } i \mid \text{prepared } j)$ [5]. For a noiseless measurement, $A_q = I$. For a noisy measurement, $A_q[1, 0] = p_{0 \rightarrow 1}$ is the probability of reading ”1” when the qubit was in $|0\rangle$ and $A_q[0, 1] = p_{1 \rightarrow 0}$ is the reverse:

$$A_q = \begin{pmatrix} 1 - p_{0 \rightarrow 1} & p_{1 \rightarrow 0} \\ p_{0 \rightarrow 1} & 1 - p_{1 \rightarrow 0} \end{pmatrix}. \quad (9)$$

The elements of A_q are obtained empirically. Two calibration circuits are run per qubit: one prepares $|0\rangle$ and measures, giving estimates of $P(0|0)$ and $P(1|0)$; the other applies an X gate to prepare $|1\rangle$ and measures, giving $P(0|1)$ and $P(1|1)$. In this study, $N_{\text{shots}}^{\text{cal}} = 2048$ shots per calibration circuit are used, yielding $\sim 0.3\%$ statistical precision on the error rates. The total calibration circuit overhead is $2N$ circuits per (N, h, L) configuration, run once and reused across all mitigation configurations and repetitions.

Given the noisy probability vector $\mathbf{p}_{\text{noisy}}$ from each measurement, the corrected probability vector is obtained by applying the inverse:

$$\mathbf{P}_{\text{corrected}} = A_q^{-1} \mathbf{P}_{\text{noisy}}. \quad (10)$$

For N qubits, the full correction would require inverting the $2^N \times 2^N$ tensor product $A = A_{N-1} \otimes \dots \otimes A_0$, which has exponential computational cost with N . Assuming that readout errors on different qubits are independent, the N -qubit correction factorises into N independent per-qubit 2×2 matrix inversions, each applied along the corresponding qubit axis of the probability tensor, reducing computational cost from $\mathcal{O}(4^N)$ to $\mathcal{O}(N \cdot 2^N)$ [15, 30]. With this method, the corrected probabilities can become negative, a known artifact of linear inversion, but the effect on the expectation value computation is negligible for the calibration precision used in this study. More advanced methods such as M3 add non-negativity constraint [31], but are outside this study's scope.

2.5.2 Parity Post-Selection

As stated in Section 2.1, the TFIM Hamiltonian commutes with the global parity operator $P = \bigotimes_{i=1}^N \sigma_i^x$ [10, 32]. This is verified by showing that each term in H commutes with P individually. The commutation relation of $P = \bigotimes_{i=1}^N \sigma_i^x$ with the $H_X = -h \sum_{i=1}^N \sigma_i^x$ term is trivial since σ_i^x commutes with itself, giving $[H_X, P] = 0$. In the ferromagnetic term, σ_i^z anticommutes with σ_i^x , such that $\sigma_i^z \sigma_i^x = -\sigma_i^x \sigma_i^z$, and σ_i^z commutes with σ_j^x for $j \neq i$. Hence, when $\sigma_i^z \sigma_{i+1}^z$ is commuted through P , both sites have a factor -1 , giving $(-1)^2 = +1$:

$$P \sigma_i^z \sigma_{i+1}^z = (-1)^2 \sigma_i^z \sigma_{i+1}^z P = \sigma_i^z \sigma_{i+1}^z P, \quad (11)$$

therefore, $[\sigma_i^z \sigma_{i+1}^z, P] = 0$ for every nearest-neighbour pair and $[H_{ZZ}, P] = 0$. Since $[H, P] = [H_{ZZ} + H_X, P] = 0$, the Hamiltonian and P share the same eigenbasis. Every energy eigenstate has a definite parity eigenvalue $p = \pm 1$. The transverse field lifts degeneracy between parity sectors, with the even-parity state having lower energy for any $h > 0$. For even N , with OBC, and any $h > 0$, the ground state of the TFIM lies in the even-parity sector $p = +1$ [7, 10]. In the computational basis, even parity corresponds to bitstrings with an even number of measured "1"s.

The parity conservation of the TFIM is exploited in the parity post-selection method. Any measurement outcome with odd parity, that is an odd number of "1"s, cannot have come from the true ground state, it must be a noise-induced error. The mitigation strategy consists of discarding all odd-parity bitstrings and renormalising the probabilities of the remaining even-parity distribution. If noise is severe enough that most outcomes have wrong parity, the sample is left with few shots and statistical uncertainty increases.

Parity post-selection has no circuit overhead: it is a purely classical post-processing routine of

existing measurement data. However, discarding readings effectively wastes a fraction of the total shots, requiring an effective shot overhead of $1/(1 - f_{\text{discard}})$, where f_{discard} is the measured discard fraction of each N, h, L combination.

2.5.3 Zero-Noise Extrapolation

The Zero-Noise extrapolation (ZNE) method operates on the assumption that $\langle H \rangle$ depends smoothly on the noise level, λ . Hence, if $\langle H \rangle$ can be measured at several controlled noise levels, a model can be fitted to extrapolate to $\lambda = 0$, the zero-noise limit [11, 33]. In a quantum device noise cannot be reduced below the hardware baseline, but it can be increased in a controlled manner.

Noise amplification by gate folding exploits the fact that each gate operation in a circuit introduces noise, as described in Section 2.4. Noise can be amplified by replacing each gate G in the circuit with the sequence:

$$G \rightarrow G(G^\dagger G)^k, \quad (12)$$

where k determines the amplification factor. Since $G^\dagger G = I$, the logic of the circuit is unchanged, but each additional gate invocation adds one application of the gate's noise channel [34]. Furthermore, the noise scale factor is:

$$\lambda = 2k + 1, \quad (13)$$

such that at $k = 0$ (no folding), $\lambda = 1$, the baseline noise; at $k = 1$, $\lambda = 3$, triple noise; and so on. Both single-qubit and two-qubit gates are folded to ensure proportional amplification of all noise sources.

Given the data points $\{(\lambda_i, E(\lambda_i))\}$ for each scale factor, a polynomial is fitted:

$$E(\lambda) = E_0 + a\lambda + b\lambda^2, \quad (14)$$

where $E_0 = E(\lambda = 0)$ is the zero-noise energy estimate recovered as the constant term. With three scale factors ($\lambda \in \{1, 3, 5\}$), a degree-2 polynomial is uniquely determined by the three data points, hence the system is exactly determined and the polynomial passes through all three measurements exactly. This procedure is known as Richardson extrapolation [35], implemented via `np.polyfit`, which reduces to exact interpolation for n points fitted with a degree $n - 1$ polynomial.

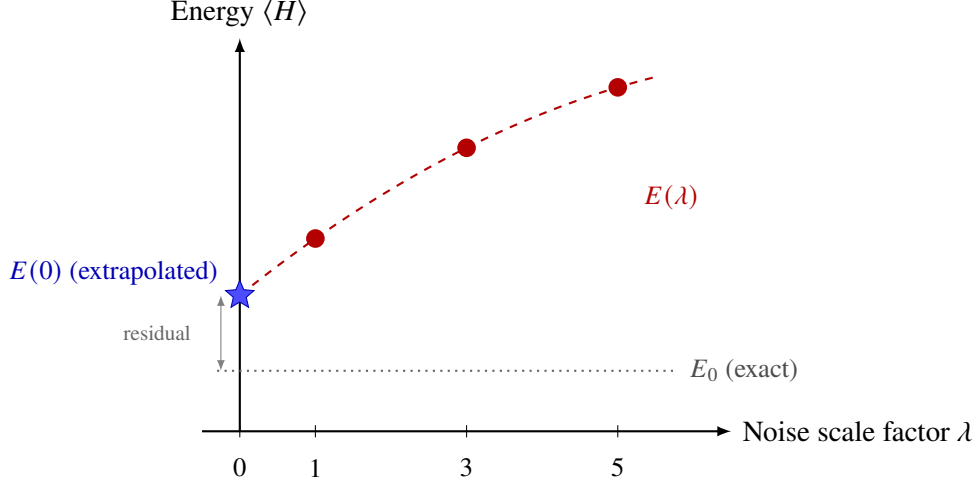


Figure 4: Schematic illustration of zero-noise extrapolation. Energy measurements at three noise scale factors $\lambda \in \{1, 3, 5\}$ (filled circles) are fitted with a quadratic polynomial (dashed curve). The extrapolated value $E(0)$ (star) estimates the zero-noise energy. The exact ground state energy E_0 is shown for reference (dotted line). The gap between $E(0)$ and E_0 represents residual error from noise sources not addressable by gate folding.

ZNE with n_λ scale factors requires n_λ pairs of additional circuit evaluations (Z-basis and X-basis) per energy estimate, one per λ . Hence, this method requires a circuit overhead of $n_\lambda \times$ relative to the unmitigated method. ZNE is therefore the most costly of the three mitigation methods in terms of circuit executions. The circuit is transpiled to the native gate set to avoid internal optimisation and ensure that each folded $G^\dagger G$ pair carries the physical noise of native gates. No further transpilation is needed, as optimisation passes would cancel the $G^\dagger G$ identity pairs. The uncertainty of the extrapolated E_0 is obtained by propagating the statistical uncertainties of the individual energy measurements at each scale factor through the polynomial fit via standard error propagation of the fitting coefficients.

2.6 Combined Mitigation Configurations

Since the error mitigation strategies described target different noise sources, and they operate at different stages of the computation pipeline, they can be applied independently or in any combination. This gives $2^3 = 8$ configurations, including the unmitigated baseline, as summarised in Table 2. This serves to identify whether combining methods produces additive, synergistic, or antagonistic effects on error reduction.

Configuration	Readout	Parity	ZNE
Raw			
R	✓		
P		✓	
Z			✓
R+P	✓	✓	
R+Z	✓		✓
P+Z		✓	✓
R+P+Z	✓	✓	✓

Table 2: Error mitigation configurations benchmarked.

When combined, the three methods must be applied in a specific sequence [21]. First, the ZNE gate folding must happen during circuit construction, before any measurements are taken. Second, readout mitigation is applied to the raw measurement counts, as it corrects the probability distribution for measurement apparatus. Third, parity post-selection is applied to the readout-corrected probability distribution. This ordering is necessary because readout bitflips can flip the parity of a bitstring, causing correct outcomes to be discarded and erroneous outcomes to be retained. Finally, the ZNE polynomial fit is applied to the expectation values obtained at each noise scale.

All mitigation configurations are applied to the final energy evaluations of $\langle H \rangle$ for each (N, h, L) , so they share the same optimised parameters θ^* . The VQE optimisation is performed once per (N, h, L) combination using the raw cost function. This ensures a fair comparison between mitigation combinations, so all differences in ϵ across configurations are due to the mitigation method alone, not differences in the variational approximation.

The key code modules to implement the error mitigation techniques described in this section are provided in Appendix C.

3 Results and Analysis

The benchmark results discussed in this section are obtained following the implementation described in Appendix A and summarised in Table A.1 with the FakeBrisbane noise model (IBM Eagle r3, 127 qubits).

To quantify the accuracy and uncertainty of VQE and each mitigation strategy, the error metrics used are the relative error with respect to ED (ϵ), defined in Equation (15), and the standard error of the mean (SEM), defined in Equation (16):

$$\epsilon = \left| \frac{E_{\text{VQE}} - E_{\text{ED}}}{E_{\text{ED}}} \right|, \quad (15)$$

where E_{VQE} is the VQE estimate of the ground state energy, and E_{ED} is the exact energy of the ground state; and

$$\text{SEM} = \frac{s}{\sqrt{n_{\text{reps}}}}, \quad (16)$$

where s is the sample standard deviation with Bessel's correction, and $n_{\text{reps}} = 10$ is the number of independent energy evaluations carried per (N, h, L) configuration. The relative error ϵ is dimensionless, scale independent, and allows meaningful comparison across different (N, h, L) configurations. The SEM captures the effect of shot noise and stochastic gate noise. All error bars on figures in this section correspond to ± 1 SEM.

COBYLA convergence was verified for all (N, h, L) configurations; convergence plots are provided in Appendix B. All curves for $N \leq 10$, $L \leq 2$ plateau well before the $\text{maxiter} = 600$ limit. At the extremal configuration $N = 12$, $L = 3$, the curves are still descending at ~ 450 iterations, justifying the chosen limit. The relative error does not always increase monotonically with N , reflecting the stochastic nature of derivative-free optimisation with $n_{\text{restarts}} = 3$ on noisy landscapes.

3.1 Individual Mitigation Methods

The most consistently beneficial individual method across all phases, system sizes and depths is readout correction, as shown in Figure 5. This method provides a $\sim 10 - 15$ percentage point reduction in ϵ over raw, at negligible additional computational expense. This method's performance shows no systematic dependence on h or L and improves monotonically with N in most cases. This result is expected: readout errors are independent of the quantum state and scale linearly with N , so per-qubit correction addresses a fixed fraction of the total readout error regardless of phase or circuit depth. Readout correction is therefore the first method to implement and the baseline above which all other methods should be evaluated.

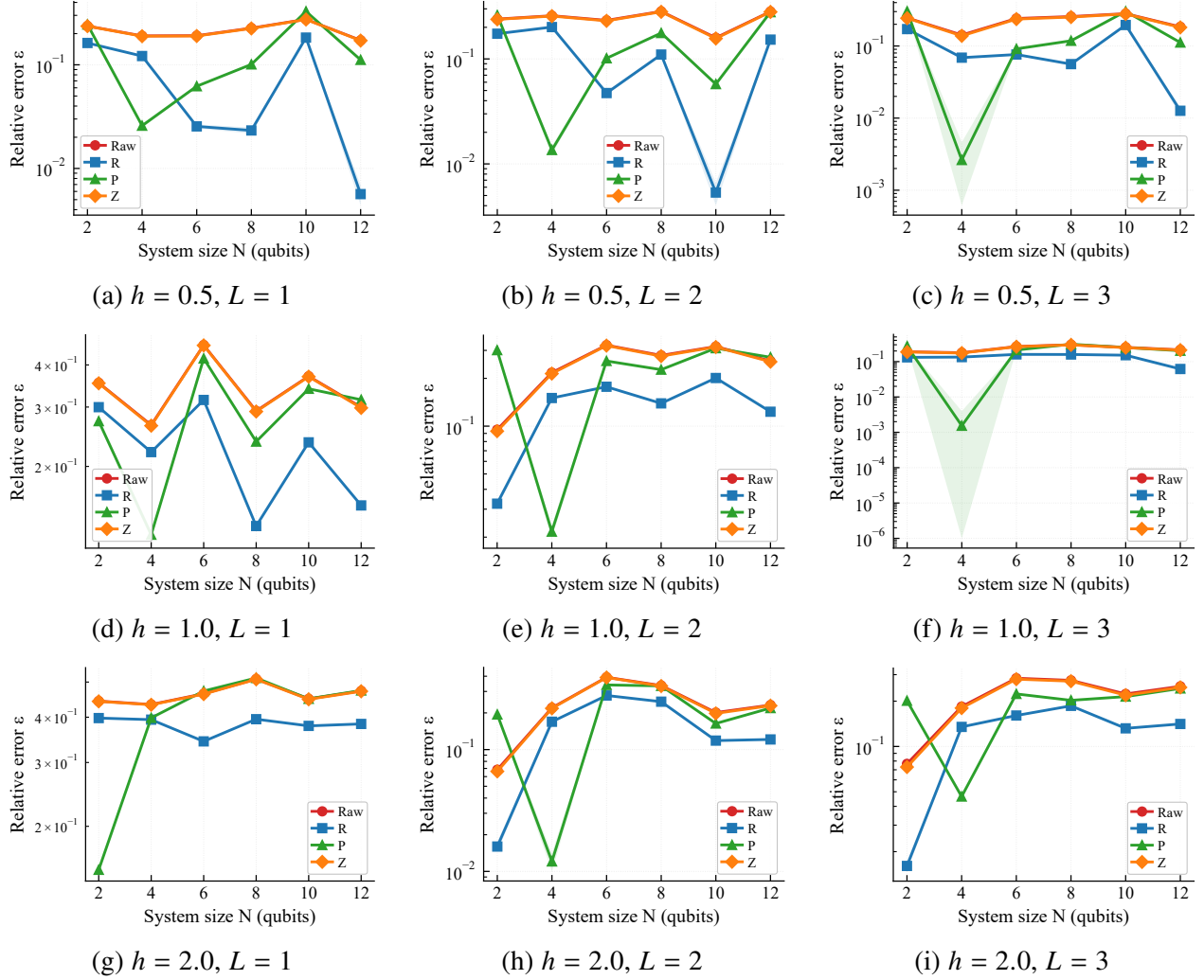


Figure 5: Relative error ϵ as a function of system size N for the four individual mitigation configurations (Raw, R, P, Z) on FakeBrisbane. Rows correspond to $h \in \{0.5, 1.0, 2.0\}$ and columns to $L \in \{1, 2, 3\}$. Shaded bands denote ± 1 SEM from $n_{\text{reps}} = 10$ independent repetitions. The ZNE (Z) and Raw curves overlap across all panels, reflecting the negligible effect of ZNE on this device.

Parity post-selection is the most phase dependent individual method [10]. In the ordered phase ($h = 0.5$) its performance is excellent, achieving near-zero relative error in some (N, h, L combinations ($\epsilon_P = 0.26\%$ at $N = 4, h = 0.5, L = 3$). The ordered ground state approximates the GHZ state $\frac{1}{\sqrt{2}}(|\uparrow \cdots \uparrow\rangle + |\downarrow \cdots \downarrow\rangle)$, which has nearly all weight in the even-parity sector [13]. VQE can find this structure reliably, so parity post-selection selectively removes genuine noise events. Discard fractions are always lowest in this regime, as shown in Figure 6.

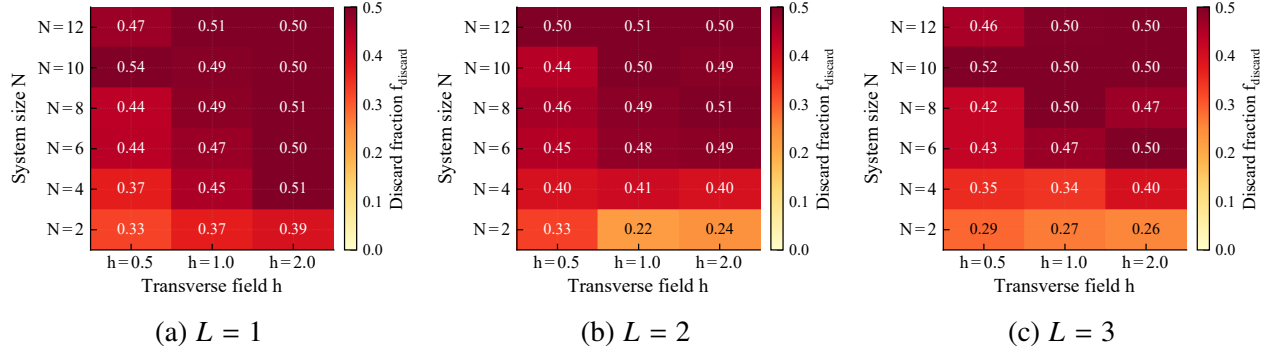


Figure 6: Measured parity discard fraction f_{discard} for the parity-only (P) configuration on FakeBrisbane. Each panel corresponds to an ansatz depth ($L = 1, 2, 3$, left to right), with system size N on the vertical axis and field strength h on the horizontal axis. Lower discard fractions (lighter colours) indicate that the VQE-prepared state has stronger overlap with the correct parity sector.

At critical and disordered phases ($h = 1.0, 2.0$) parity post-selection is unreliable, sometimes only marginally different to raw ($\epsilon_{\text{raw}} = 46\%$ and $\epsilon_P = 42\%$ at $N = 6$, $h = 1.0$, $L = 1$). The critical ground state is a complex entangled superposition that shallow noisy circuits approximate poorly due to a lack of expressibility, and the approximate state does not sit cleanly in either parity sector. Hence, parity post-selection discards near-randomly without selective error removal. In the disordered phase ($h = 2$), parity is near-useless for $L = 1$. At $L = 2, 3$ the situation partially recovers as VQE finds states with more parity structure.

The measured discard fractions increase with N and approach 0.5 asymptotically. Anomalously low discards at $N = 2$, $L \geq 2$, $h = 1.0$ ($f_{\text{discard}} \approx 0.22 - 0.27$) can be seen, and are interpreted as VQE successfully preparing a state close to the true even-parity critical ground state. This confirms that parity post-selection requires the VQE circuit to prepare a state with the correct symmetry structure, making its effectiveness dependent on ansatz expressibility and optimiser convergence.

ZNE provides virtually no measurable improvement over raw across all (N, h, L) combinations on FakeBrisbane. In Figure 5, the ZNE and raw curves are visually indistinguishable. This is not an implementation error, as diagnostic checks confirm that gates are folded correctly and $E(\lambda = 1, 3, 5)$ shows the correct monotonic increase with λ . The failure is physical: on FakeBrisbane, readout errors (2.51%) dominate gate errors (0.78%) by a factor of 3.2 $\times$. ZNE via gate folding targets gate errors exclusively, whereas readout errors occur at measurement and are invisible to the folding procedure [36]. Hence, after Richardson extrapolation removes the gate-error component, the dominant readout bias remains [11]. The ZNE null result motivates the noise profile study in Section 4.

The dependence of error mitigation performance on system size $N \in \{2, 4, 6, 8, 10, 12\}$ is mild throughout. Relative error generally increases with N for all methods, consistent with the accumulation of per-gate and per-qubit errors across more qubits and deeper transpiled circuits. This is most

clearly seen at $h = 2.0$, $L = 2, 3$.

3.2 Combined Mitigation Strategies

The combination of readout correction plus parity post-selection (R+P) is the most effective method in the ordered phase, and it remains competitive at criticality, though the triple combination R+P+Z achieves marginally lower error at $L = 3$, as shown in Figure 7. The readout correction cleans the probability distribution from measurement-induced bitflips, then parity post-selection operates on corrected data and selectively removes true gate-error events. The combined error reduction exceeds the sum of individual reductions, indicating that the two methods address non-overlapping error populations, as shown in Figure 8.

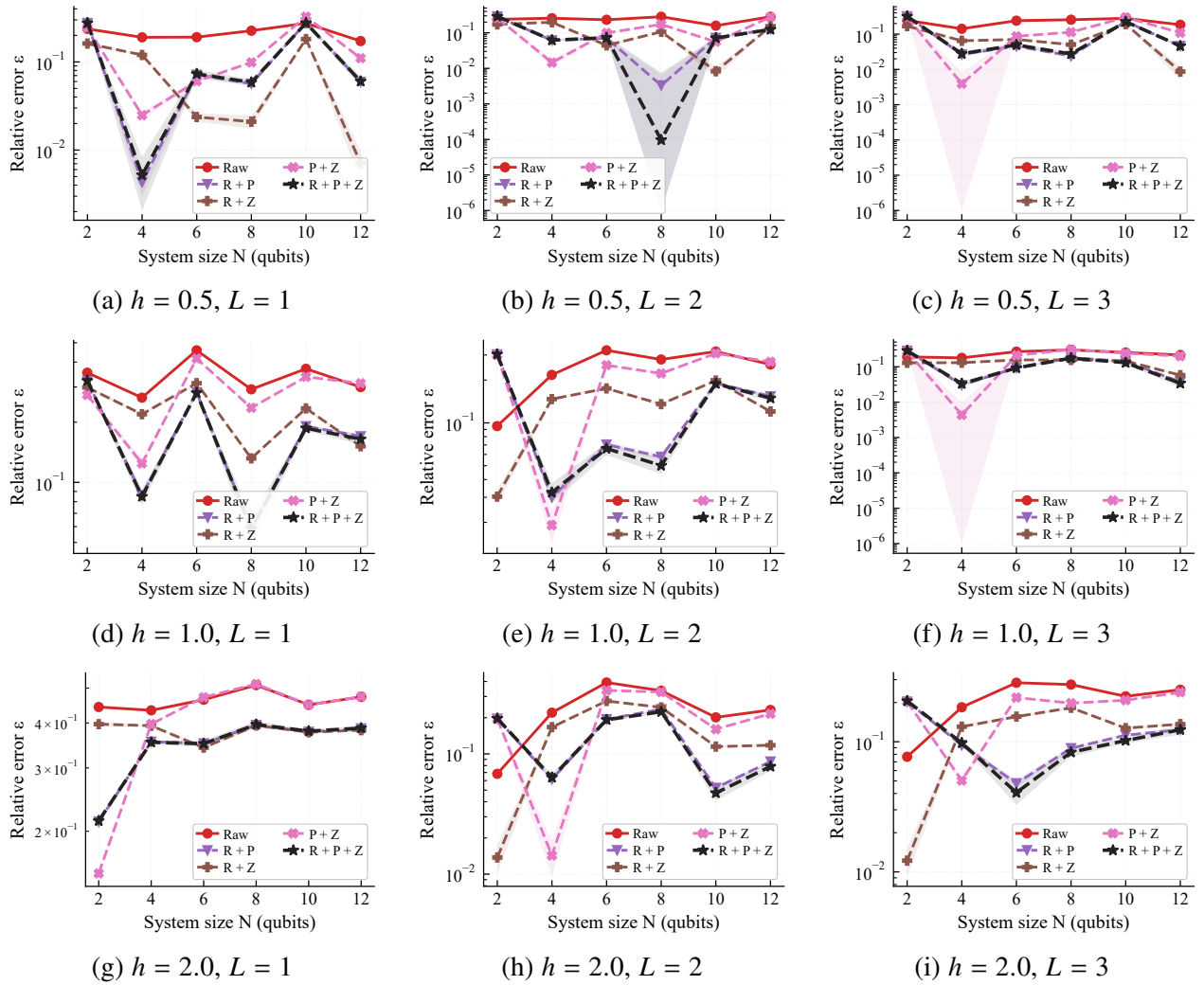


Figure 7: Relative error ϵ as a function of N for the four combined mitigation configurations (R+P, R+Z, P+Z, R+P+Z) on FakeBrisbane, with the same panel layout as Figure 5. Shaded bands denote ± 1 SEM.

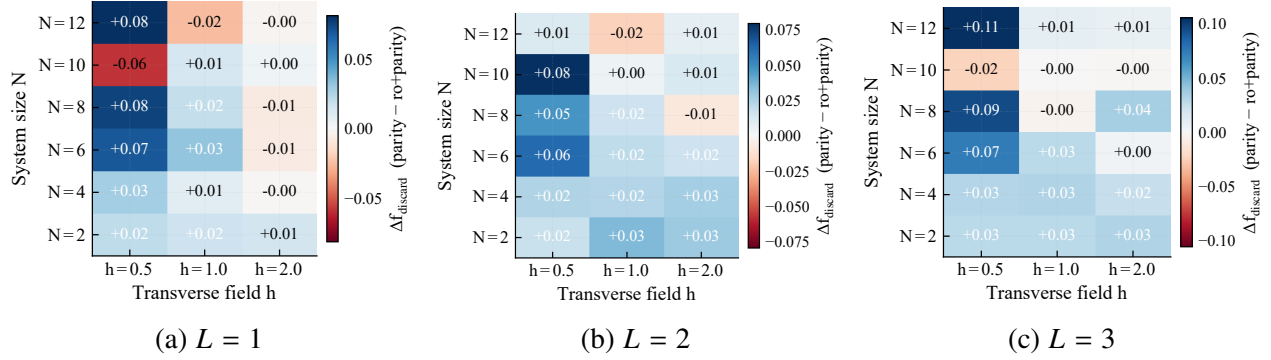


Figure 8: Reduction in parity discard fraction when readout correction is applied before parity post-selection, $\Delta f = f_{\text{discard}}^{(P)} - f_{\text{discard}}^{(R+P)}$, on FakeBrisbane. The small magnitude (1–2 percentage points) confirms that parity discards are dominated by gate errors rather than readout errors.

In the disordered phase, however, parity post-selection discards $\sim 50\%$ of shots without providing any discriminating power, making it ineffective. Readout correction combined with ZNE provides the largest error reduction in this regime, as ZNE addresses the gate-error component, however small this may be.

The full triple combination (R+P+Z) does not achieve a lower error than R+P in the ordered phase or lower error than R+Z in the disordered phase, and it marginally does in the critical phase. On FakeBrisbane’s readout-dominated noise profile, ZNE’s marginal contribution is insufficient to justify the additional circuit executions. This result is specific to Eagle r3, the comparison with FakeFez discussed in Section 5.2 shows that R+P+Z becomes optimal on hardware with a more balanced noise profile.

The dependence of error mitigation performance on ansatz depth from $L = 1$ to $L = 3$ is largest in the disordered phase, reflecting a higher expressibility requirement. At $L = 1$, $h = 2.0$, all methods plateau at $\epsilon \approx 40 - 50\%$ error regardless of N because the paramagnetic ground state is inexpressible by a single RY+CZ layer. This plateau represents the ansatz approximation error floor — the minimum error achievable by any mitigation strategy at this depth — and is different from the noise-induced error that mitigation methods address. The two contributions are not separated quantitatively in this study, though the sharp improvement from $L = 1$ to $L = 3$ at $h = 2.0$ confirms that the $L = 1$ error is dominated by ansatz expressibility rather than hardware noise. At $L = 3$ the results improve substantially, and mitigation methods begin to show meaningful differentiation.

3.3 Cost–Error Trade–Off

The Pareto frontier is a concept from multi-objective optimisation: it identifies the set of configurations for which no alternative achieves both lower error, ϵ , and lower cost, Ω_{eff} , simultaneously, rendering

them as Pareto-optimal. Configurations on the frontier represent the best available trade-offs, and those inside the frontier are dominated. The Pareto frontier, shown in Figure 9, shows the median relative error ϵ of each error mitigation technique as a function of its effective circuit overhead, that is, the extra number of quantum circuits it requires, which is used as a cost metric [37]. It is defined as $\Omega_{\text{eff}} = \frac{n_\lambda}{1-f_{\text{discard}}} + \frac{2N}{N_{\text{all eval}}}$, where $n_\lambda = 3$ for ZNE configurations and $n_\lambda = 1$ for non-ZNE configurations, and f_{discard} is the measured parity discard fraction [8]. The readout calibration circuits add $2N$ circuit overhead amortised across all evaluations, contributing $\lesssim 5\%$ additional overhead. Hence, the dominant cost is $\frac{n_\lambda}{1-f_{\text{discard}}}$. Alternative cost metrics — such as total wall-clock time, cumulative shot count including calibration, or classical post-processing overhead — could also be used. The effective circuit overhead was chosen because it captures the dominant quantum resource scaling and is independent of classical hardware. Comparing Figure 9a, Figure 9b, and Figure 9c, it can be seen that the Pareto frontier is phase dependent.

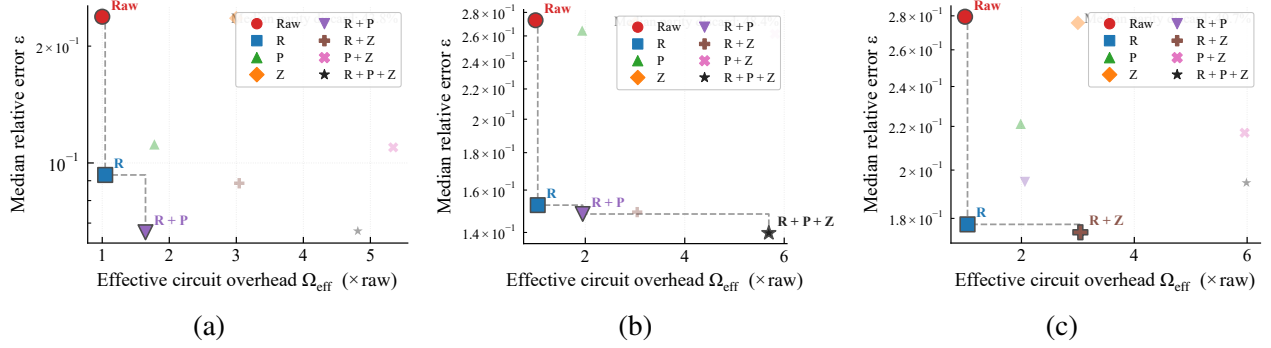


Figure 9: Cost–error Pareto frontier on FakeBrisbane for (a) ordered ($h = 0.5$), (b) critical ($h = 1.0$), and (c) disordered ($h = 2.0$) phases. Each point represents the median ϵ and effective overhead Ω_{eff} of a mitigation configuration across all (N, L) combinations. Connected points form the Pareto frontier; configurations inside the frontier are dominated.

Phase	Pareto frontier
Ordered ($h = 0.5$)	Raw $\rightarrow$ R $\rightarrow$ R+P
Critical ($h = 1.0$)	Raw $\rightarrow$ R $\rightarrow$ R+P $\rightarrow$ R+P+Z
Disordered ($h = 2.0$)	Raw $\rightarrow$ R $\rightarrow$ R+Z

Table 3: Pareto-optimal mitigation configurations on FakeBrisbane, using the abbreviations defined in Table 2. The triple combination appears only marginally in the critical phase.

On Fake Brisbane, the Pareto frontier is summarised in Table 3. In the ordered phase, ZNE-containing methods do not appear. In the critical phase, the triple combination appears marginally. In the disordered phase, parity-based methods do not appear due to their high discard fraction, which significantly increase the effective circuit overhead.

This phase dependence of the Pareto frontier reflects the underlying physics: parity post-selection exploits the $\mathbb{Z}_2$ symmetry structure of the ground state, which is approximated best in the ordered phase and worse in the disordered phase. Readout correction is phase-independent and forms the anchor of every frontier. ZNE is uniformly weak on the Brisbane device, but becomes relevant in the disordered phase where parity fails. These results demonstrate that selecting the optimal error mitigation strategy requires knowledge of the target Hamiltonian’s symmetry and the device’s noise composition, a conclusion further supported by the noise composition study in Section 4 and the results in Section 5.

4 ZNE Noise Composition Study

The ZNE null result observed on FakeBrisbane in Section 3.1 raises the question of whether ZNE’s ineffectiveness is fundamental or specific to the noise composition of the Brisbane processor [8]. This section presents a parametric study that isolates the role of the noise ratio $r = p_{\text{gate}}/p_{\text{readout}}$ (as defined in Section 2.4) in determining ZNE performance. A synthetic noise model with independently tunable gate and readout error probabilities is constructed and used to evaluate ZNE performance across a range of noise profile compositions. Because ZNE via gate-folding targets gate noise exclusively, readout errors are not addressed by this method [36]. This separability allows ZNE effectiveness to be studied as a function of noise composition independently of other device-specific factors.

4.1 Synthetic Noise Model Design

A custom synthetic noise model is constructed in AerSimulator with two independent components: (i) depolarising channels applied after every single-qubit and two-qubit gate, with error probability p_{gate} , and (ii) readout bit-flip errors applied at measurement, with symmetric error probability p_{readout} , such that $p_{0 \rightarrow 1} = p_{1 \rightarrow 0} = p_{\text{readout}}$. The total noise budget is fixed at $p_{\text{total}} = p_{\text{gate}} + p_{\text{readout}} = 0.033$, chosen to match the approximate total error rate of FakeBrisbane. The noise ratio is then $r = p_{\text{gate}}/p_{\text{readout}}$, which is varied while keeping p_{total} constant. The synthetic p_{gate} represents a single depolarising rate applied uniformly to all gate operations, serving as a proxy for the dominant two-qubit gate contribution. When placing real devices on the r -curve in Section 4.3, r is computed from the two-qubit gate error rate and per-qubit readout error rate from Table 1, as two-qubit gate errors dominate the gate-error budget for the shallow circuits in this study.

The VQE parameters θ^* in this study are obtained from noiseless optimisation (ideal simulator, no gate or readout noise), making them always near-optimal. This decouples the results from COBYLA convergence quality, ensuring that any variation in ϵ is attributed solely to the noise model

composition. The circuit parameters are fixed at $N = 4$, $h = 1.0$, and $L = 2$ (critical phase, moderate depth), the regime where ZNE has the most room to contribute.

The metric used to quantify ZNE effectiveness is ZNE improvement fraction, defined as

$$\eta_{\text{ZNE}} = \frac{\epsilon_{\text{raw}} - \epsilon_{\text{ZNE}}}{\epsilon_{\text{raw}}}, \quad (17)$$

where ϵ_{raw} is the relative error without ZNE and ϵ_{ZNE} is the relative error after Richardson extrapolation. Both raw and ZNE evaluations use the same θ^* ensuring a fair comparison. $\eta_{\text{ZNE}} = 0$ means ZNE provides no improvement, $\eta_{\text{ZNE}} = 1$ would mean ZNE eliminates all error. Negative values would indicate ZNE worsens accuracy. The expected effect of noise composition on ZNE is illustrated schematically in Figure 10.

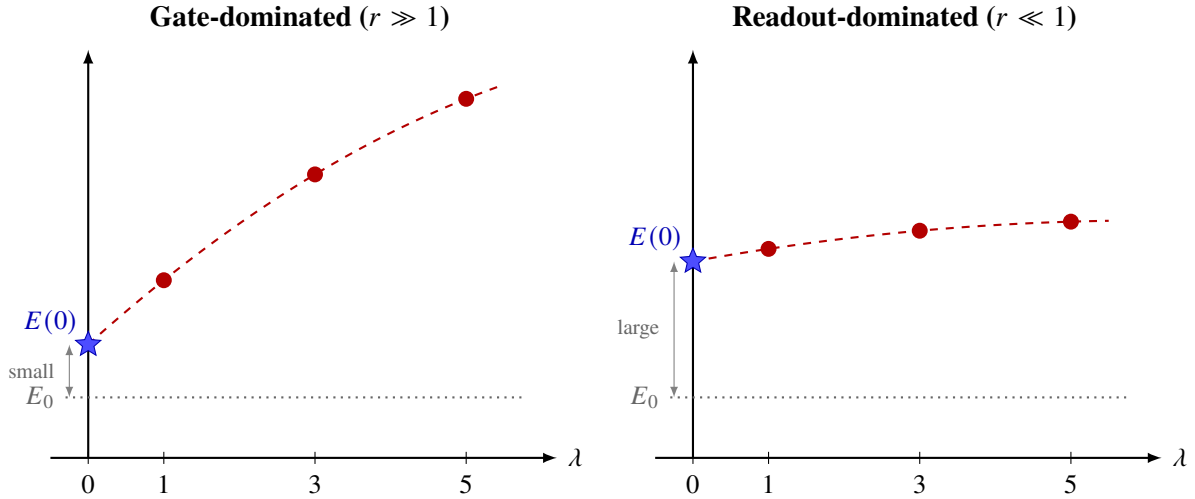


Figure 10: Schematic comparison of zero-noise extrapolation on two noise profiles. **Left:** when gate errors dominate ($r \gg 1$), the energy increases steeply with λ , producing a large Richardson signal and placing $E(0)$ close to E_0 . **Right:** when readout errors dominate ($r \ll 1$), the energy depends weakly on λ because gate folding does not amplify readout noise. The extrapolated $E(0)$ remains far from E_0 , with the residual dominated by readout bias that is invisible to the folding procedure. This contrast is quantified by the continuous ratio sweep in Section 4.3.

4.2 Discrete Models

Figure 11 shows three pairs of bars. Each pair represents the ϵ_{raw} and ϵ_{ZNE} obtained in the synthetic noise models.

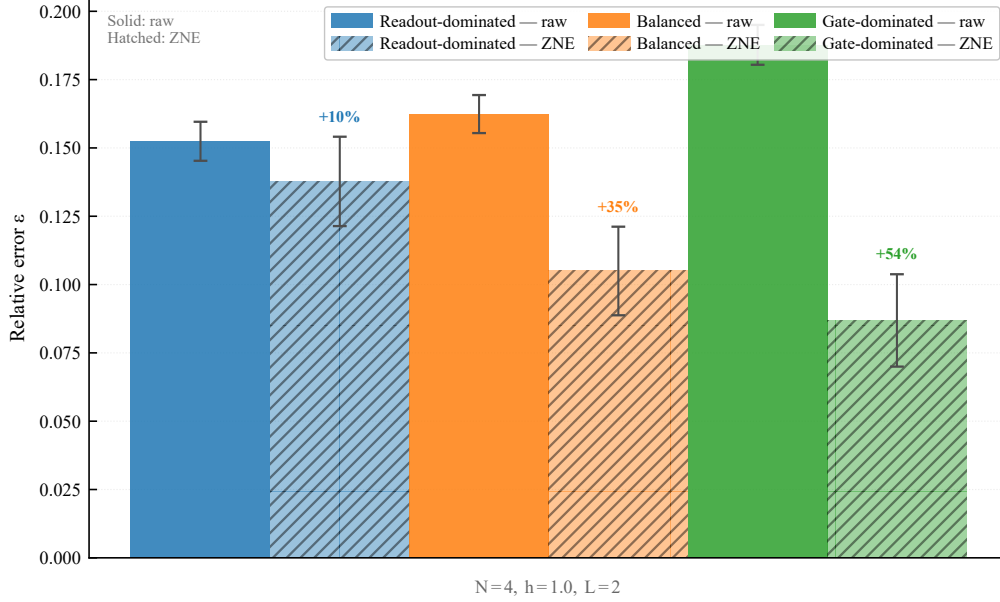


Figure 11: Relative error ϵ before (solid) and after (hatched) ZNE for three synthetic noise models at fixed total noise $p_{\text{total}} = 0.033$: readout-dominated ($r \approx 0.1$), balanced ($r = 1$), and gate-dominated ($r \approx 10$). Circuit parameters: $N = 4$, $h = 1.0$, $L = 2$. Error bars show ± 1 SEM. The ZNE improvement fraction η_{ZNE} increases from 10% to 54% as the gate-error contribution grows.

In the readout-dominated model in Figure 11 ($r = 0.1$), ZNE provides only $\eta_{\text{ZNE}} = 10\%$ improvement. This value is small because gate errors contribute a small fraction of the total noise. As mentioned in Section 3.1, after removing the gate error, the dominating readout bias remains.

At $r = 1$, gate and readout errors contribute comparably, and ZNE achieves $\eta_{\text{ZNE}} = 35\%$, showing substantial improvement and making ZNE effective in this regime. The residual error is attributed to the readout component, which remains uncorrected.

The gate-dominated model ($r = 10$) shows $\eta_{\text{ZNE}} = 54\%$ because ZNE is correcting the dominating error channel, providing and placing $E(0)$ closer to E_0 . However, η_{ZNE} does not reach 100% because Richardson extrapolation with a finite degree polynomial cannot perfectly capture the nonlinear scaling of depolarising noise under gate folding, leaving a residual bias even when gate errors dominate. On real devices, T_1/T_2 decoherence would impose an additional, independent saturation mechanism.

The three discrete models demonstrate that ZNE effectiveness increases monotonically with r . The η_{ZNE} transitions from negligible in the readout-dominated regime to substantial in the gate-dominated regime. Hence, the physical mechanism becomes clear: gate folding can only amplify, and therefore allow extrapolation to, the gate-error contribution of the total noise. When this component is a small fraction of the total noise (low r), ZNE can only mitigate that small fraction, at best.

4.3 Continuous Ratio Sweep

Ten noise models are constructed at log-spaced values of r from 0.05 to 20, all at fixed $p_{\text{total}} = 0.033$. For each model, η_{ZNE} is computed at $N = 4$, $h = 1.0$, $L = 2$ using the same noiseless optimal parameters θ^* . Figure 12 shows η_{ZNE} as a function of r .

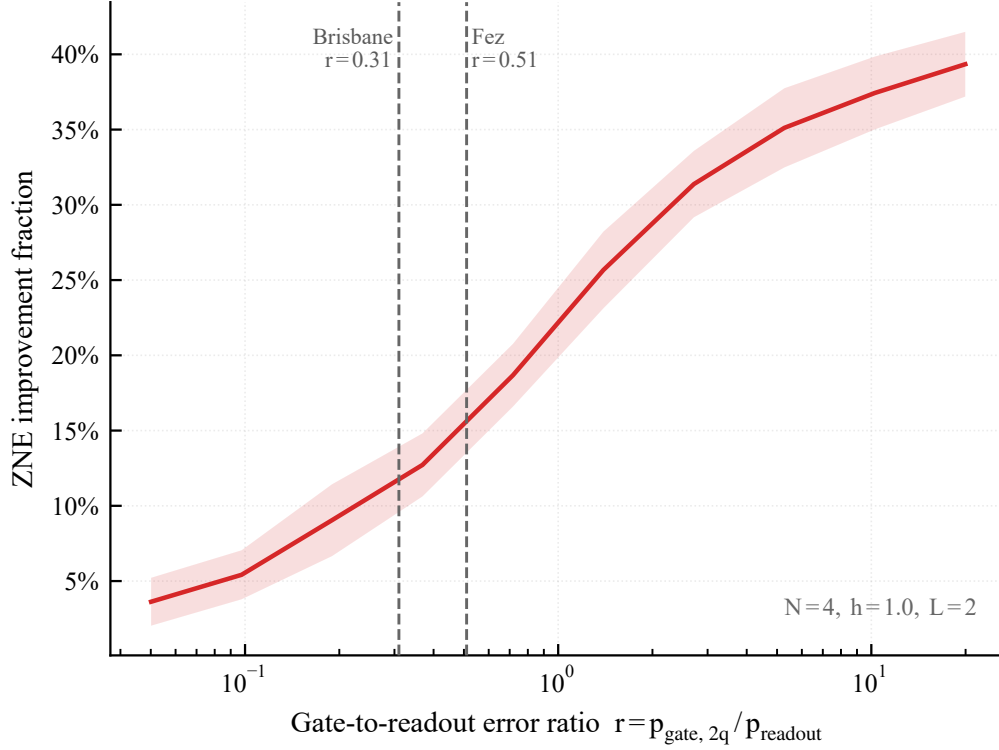


Figure 12: ZNE improvement fraction $\eta_{\text{ZNE}} = (\epsilon_{\text{raw}} - \epsilon_{\text{ZNE}}) / \epsilon_{\text{raw}}$ as a function of the gate-to-readout noise ratio $r = p_{\text{gate}} / p_{\text{readout}}$ at fixed $p_{\text{total}} = 0.033$, $N = 4$, $h = 1.0$, $L = 2$. The shaded region show ± 1 SEM. The dashed lines marks $r_{\text{Brisbane}} = 0.31$ and $r_{\text{FeZ}} = 0.51$. The curve exhibits sigmoidal dependence, saturating at $\sim 40\%$ due to finite-degree polynomial approximation inherent in Richardson extrapolation.

As seen in Figure 12, $\eta_{\text{ZNE}} \geq 0$ across the entire range, meaning that ZNE never worsens the energy estimates. This shows that Richardson extrapolation does not introduce systematic bias in any noise regime tested. Additionally, the η_{ZNE} curve has a sigmoidal dependence on r . At low r ($\lesssim 0.1$), ZNE improvement is minimal and approximately constant at $\sim 4\%$, since gate errors constitute a small fraction of the total noise. In the transition region ($r \approx 0.2 - 2$), η_{ZNE} rises steeply as the gate-error fraction of the total budget increases. At high r ($\gtrsim 5$), the improvement saturates at $\sim 40 - 42\%$, never approaching 100%. This saturation arises because the polynomial Richardson model assumes linear scaling of noise with the scale factor λ , which is only approximately satisfied by depolarising channels at high fold factors [34]. On real hardware, T_1/T_2 decoherence provides an additional saturation mechanism: as gate folding increases, circuit duration accumulates decoherence that the

polynomial model cannot capture [11, 38]. Lastly, the crossover r^* where ZNE begins providing meaningful benefit ($\eta_{\text{ZNE}} \gtrsim 15\%$) is around $r \approx 0.4 - 0.5$.

4.4 Noise Profile On Real Devices

The FakeBrisbane noise profile ($r = 0.31$) places it in the readout-dominated regime, where the η_{ZNE} curve predicts $\sim 10\%$ improvement. This observation is consistent with the near-zero visible benefit observed in Section 3.1. IBM’s Fez processor sits at $r = 0.51$, predicting $\sim 20\%$ improvement, which is validated by the FakeFez benchmark results presented in Section 5.2.

The parametric study serves as a predictive framework: given a device’s noise characterisation, which is often available from calibration data, one can estimate whether ZNE will be effective before running the full benchmark. The FakeFez improvement, shown in Section 5.2, was predicted by the r -curve before the benchmark was run, providing independent validation of the parametric model. This result demonstrates that ZNE effectiveness is not a fixed property of the algorithm but depends critically on the noise profile of the underlying hardware [12, 39].

The dependence of ZNE effectiveness on noise composition has been noted independently by Sobrino et al. [12], who characterise it through a qubit error probability metric that aggregates gate and readout contributions into a single per-qubit quantity. The framework presented here differs in isolating the gate-to-readout ratio r as the controlling variable at fixed total noise, which makes the underlying mechanism explicit: gate folding amplifies only the gate-error component, so ZNE effectiveness is governed by that component’s fraction of the total budget. Crucially, because r is computable directly from standard calibration data, the resulting curve is predictive across device architectures, as demonstrated by the Eagle r3 to Heron r2 comparison. The two approaches are consistent and complementary: both establish that ZNE effectiveness is a property of the hardware noise structure rather than of the algorithm itself.

5 FakeFez Comparison and Hardware Validation

This section presents results from the FakeFez noise model (IBM Heron r2) and validates the simulation methodology against real hardware runs on `ibm_fez`. The FakeFez benchmark was motivated by hardware access constraints: cloud access is available for `ibm_fez` but not for `ibm_brisbane`, and it provides a valuable comparison between two processor architectures with different noise profiles. The noise composition differences between Eagle r3 and Heron r2 were introduced in Section 2.4, Table 1.

5.1 Noise Model Differences (Eagle r3 vs Heron r2)

The Eagle r3 processor (Brisbane) uses the ECR (Echoed Cross-Resonance) native two-qubit gate, which carries both incoherent depolarising and coherent error components. Coherent errors do not satisfy the polynomial noise model assumed by ZNE, potentially reducing Richardson extrapolation accuracy. Heron r2 uses a CZ gate implemented via tunable couplers, producing more purely incoherent (depolarising) noise that better satisfies ZNE’s assumptions. Additionally, Heron r2 incorporates TLS (two-level system) mitigation, suppressing non-Markovian noise that would break the polynomial Richardson model [15, 28, 40]. Due to these improvements, Heron r2 represents a transition beyond the traditional NISQ regime towards utility-scale quantum computing, although the error mitigation techniques studied in this work remain applicable in both regimes [40].

The main quantitative difference of the architecture change is that readout error is $3.3\times$ lower on Fez (0.76% vs 2.51%). This shifts the noise ratio from $r = 0.31$ on Brisbane (readout-dominated) to $r = 0.51$ on Fez (more balanced). According to the parametric study in Section 4, this shift predicts that ZNE should transition from ineffective to moderately effective, since the gate-error fraction of the total error budget is increased.

5.2 FakeFez Benchmark Results

The benchmark results discussed in this section are obtained following the same implementation as in Section 3, described in Appendix A. In this section, however, the FakeFez noise model (IBM Heron r2, 156 qubits) was used.

The central finding in the FakeFez benchmark is that ZNE provides a visible and consistent improvement over raw, as seen in Figure 13, in contrast to its null result in FakeBrisbane. The ZNE curve sits clearly below the raw curve across most (N, h, L) combinations. The error reduction is small at $L = 1$, moderate at $L = 2$, and largest at $L = 3$. This L -dependence is precisely what ZNE theory predicts: more layers produce more CZ gates, more gates to fold, and a larger Richardson signal. The observed improvement of $\sim 15 - 20\%$ is consistent with the $r = 0.51$ prediction from the parametric study in Section 4.

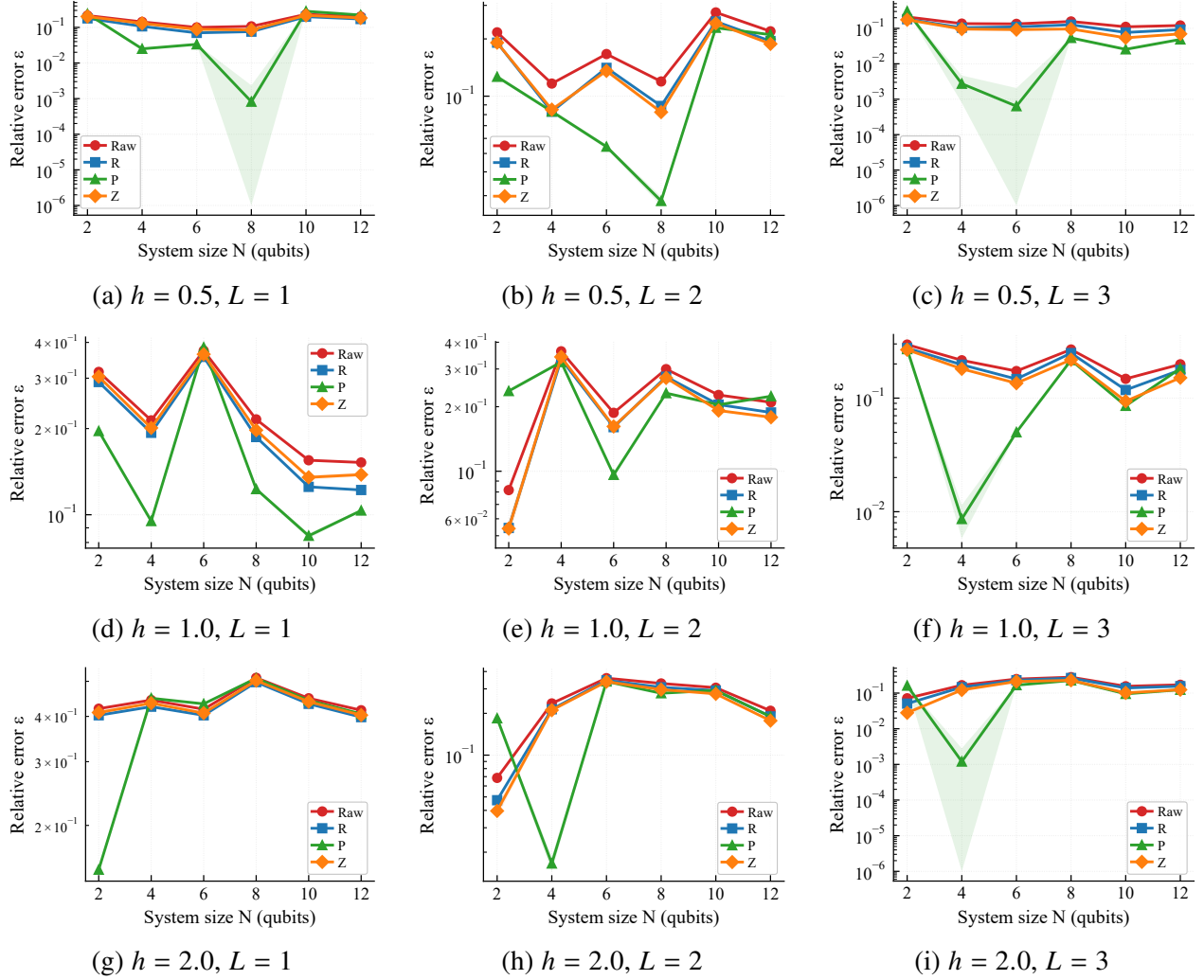


Figure 13: Relative error ϵ as a function of N for the four individual mitigation configurations on FakeFez (IBM Heron r2), with the same panel layout as Figure 5. Unlike FakeBrisbane, the ZNE curve sits visibly below the raw curve, consistent with the more balanced noise ratio $r = 0.51$.

On FakeFez, readout correction remains the most robust individual method, consistent with FakeBrisbane, but with smaller absolute error reduction due to the lower readout error rate in the Heron r2 device. Parity post-selection shows similar phase dependence as on FakeBrisbane: discard patterns are broadly consistent, with the anomalously low discards at $N = 2$, $L \geq 2$, $h = 1.0$ ($f_{\text{discard}} \approx 0.22 - 0.28$) persisting, confirming these as a physical signal independent of noise model. As shown in Figure 14, applying readout correction before parity reduces discards by only 1 – 2 percentage points on FakeFez, similar to FakeBrisbane, confirming that parity discards are gate-noise dominated.

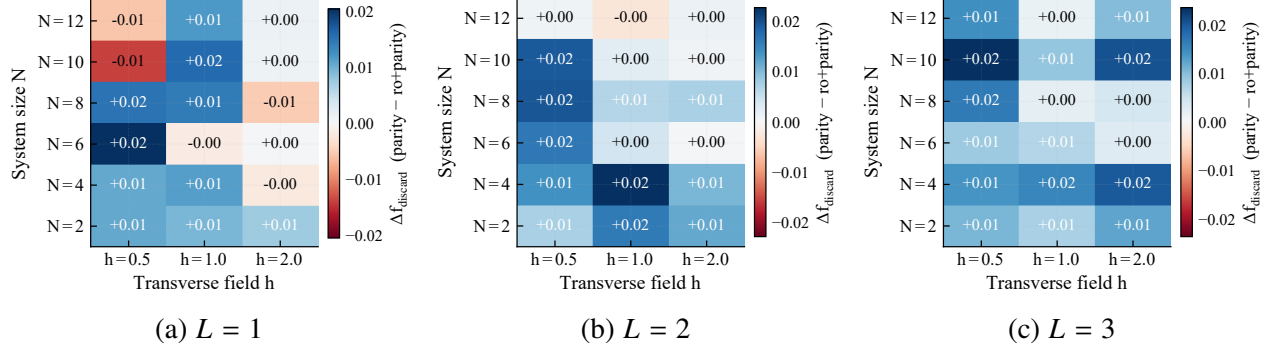


Figure 14: Reduction in parity discard fraction Δf when readout correction precedes parity post-selection on FakeFez. The pattern is consistent with FakeBrisbane (Figure 8), confirming that parity discards are gate-noise dominated on both devices.

The Pareto frontier shows important changes on FakeFez, shown in Figure 15 and Table 4. ZNE-containing methods now appear on the frontier in all three phases. Additionally, the triple combination R+P+Z sits marginally outside the frontier in the disordered phase. The phase dependence of the frontier persists, but the composition shifts, the triple combination R+P+Z becomes Pareto-optimal at $h = 1.0$, reflecting that all three methods contribute meaningfully when the noise profile is more balanced.

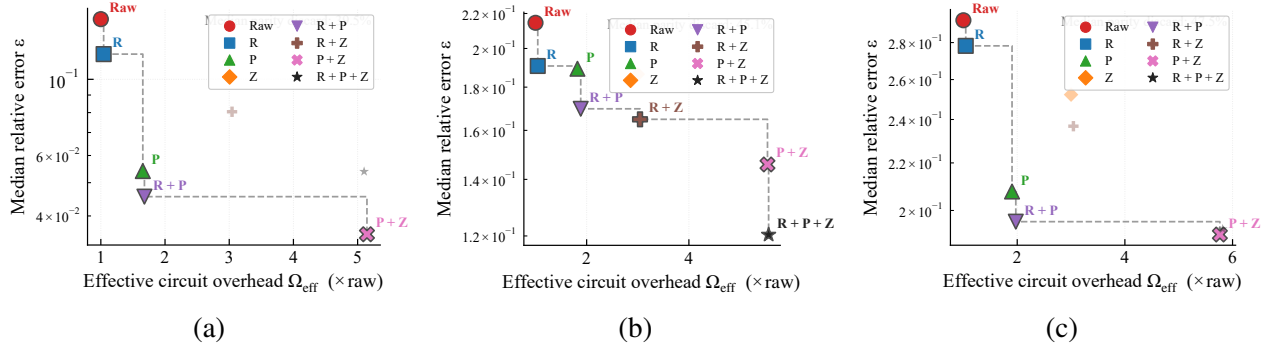


Figure 15: Cost-error Pareto frontier on FakeFez for (a) ordered, (b) critical, and (c) disordered phases. ZNE-containing methods now appear on the frontier in all three phases, and the triple combination R+P+Z is Pareto-optimal at criticality, a result that never occurs on FakeBrisbane.

Phase	Pareto frontier
Ordered ($h = 0.5$)	Raw $\rightarrow$ R $\rightarrow$ P $\rightarrow$ R+P $\rightarrow$ P+Z
Critical ($h = 1.0$)	Raw $\rightarrow$ R $\rightarrow$ P $\rightarrow$ R+P $\rightarrow$ R+Z $\rightarrow$ P+Z $\rightarrow$ R+P+Z
Disordered ($h = 2.0$)	Raw $\rightarrow$ R $\rightarrow$ P $\rightarrow$ R+P $\rightarrow$ P+Z

Table 4: Pareto-optimal mitigation configurations on FakeFez, using the abbreviations defined in Table 2. ZNE-containing methods appear in all three phases, and the triple combination is Pareto-optimal at criticality.

COBYLA convergence curves extend further on Fez than they do on Brisbane. Since the optimisation is performed on the noisy simulator, the lower readout noise provides more accurate function evaluations, allowing the optimiser to make better progress. This is most visible at $L = 3$ where some $N = 12$ curves are still descending at iteration 500.

5.3 Real Hardware Validation

Two experiments were conducted on real `ibm.fez` hardware at $N = 4$, $h = 1.0$, $L = 2$. This configuration was chosen because it is computationally manageable, lies in the critical phase where mitigation has the most leverage, and has $L = 2$ depth, where ZNE is expected to show moderate error reduction on Heron r2. Since the VQE algorithm is too costly to run on real hardware with limited cloud access, the optimal parameters θ^* were reused from the FakeFez benchmark run for the same (N, h, L) combination. $N_{\text{shots}}^{\text{eval}} = 8192$ evaluation shots, $N_{\text{shots}}^{\text{cal}} = 4096$ calibration shots and $n_{\text{reps}} = 10$ repetitions were used for this study.

Figure 16 shows $E(\lambda)$ as a function of the scale factor λ . $E(\lambda)$ increases monotonically from $E(\lambda = 1) = -3.5$ to $E(\lambda = 5) = -1.9$, confirming that gate folding correctly amplifies noise on real hardware. The Richardson extrapolation yields $E(\lambda = 0) \approx -4.3$, compared to the exact value $E_0 = -4.8$. The extrapolated value represents an improvement over $E(\lambda = 1)$, reducing ϵ from $\sim 26\%$ to $\sim 10\%$, similar to the ZNE improvement fraction predicted for this processor in Section 4.4.

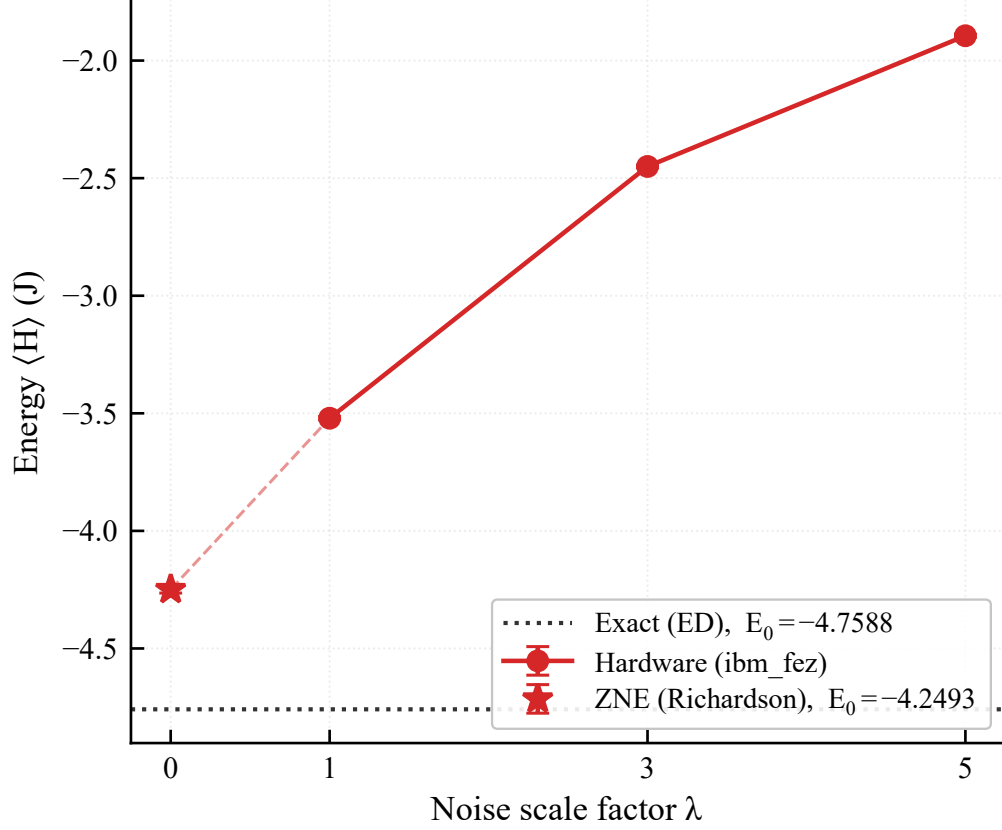


Figure 16: Energy $\langle H \rangle$ measured on real `ibm_fez` hardware as a function of the ZNE noise scale factor λ at $N = 4$, $h = 1.0$, $L = 2$. The monotonic increase from $E(\lambda = 1) \approx -3.5$ to $E(\lambda = 5) \approx -1.9$ confirms correct gate-folding behaviour. The Richardson-extrapolated value $E(0) = -4.3$ reduces ϵ from $\sim 26\%$ to $\sim 10\%$ relative to the exact energy $E_0 = -4.8$.

A single (N, h, L) combination mitigation benchmark was also carried out to compare the FakeFez simulated results to the real hardware. As shown in Figure 17, the FakeFez simulation predictions closely match the hardware results across all eight configurations. For example, the raw configuration gives $\epsilon \approx 0.36$ on FakeFez and $\epsilon \approx 0.40$ on hardware, while R+Z gives $\epsilon \approx 0.31$ on both. The best performing methods on the Fez device are parity+ZNE and the triple combination readout+parity+ZNE, consistent with the FakeFez simulation predictions. This comparison supports FakeFez as a faithful noise model for `ibm_fez` and suggests that the simulation-based conclusions in Section 5.2 are representative of real device behaviour.

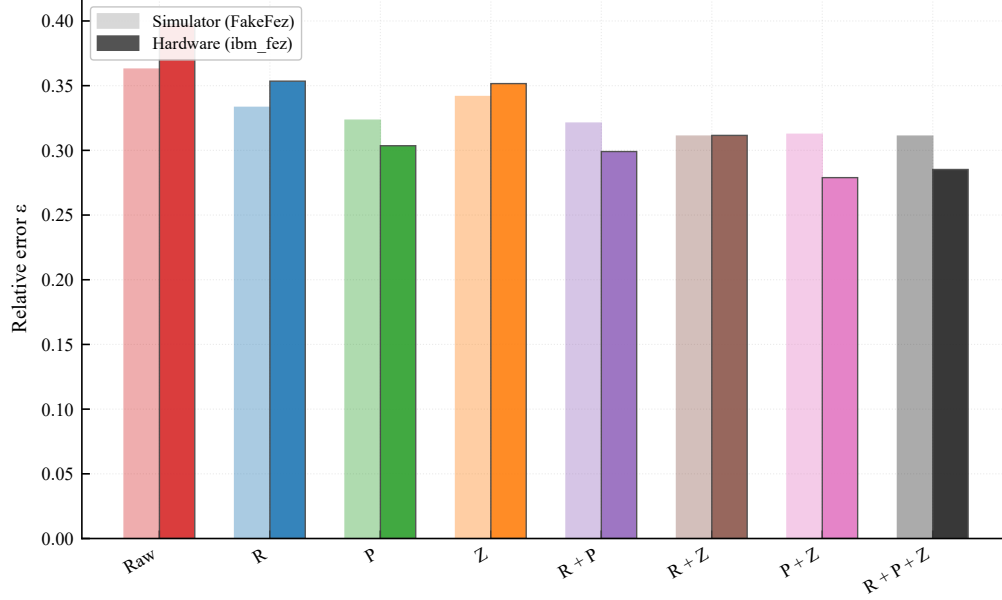


Figure 17: Comparison of relative error ϵ between real `ibm_fez` hardware (dark bars) and FakeFez simulation (light bars) across all eight mitigation configurations at $N = 4$, $h = 1.0$, $L = 2$. The close agreement validates FakeFez as a faithful noise model for the real device.

Because only one (N, h, L) combination was run on quantum hardware, the full phase and size dependencies are not validated with these tests. Furthermore, real hardware noise is dynamic, hence running the same measurements on Fez hardware would yield slightly different results each time, while the calibration data used by FakeFez is an instantaneous, constant snapshot. The parameters θ^* were optimised on FakeFez, not on the real Fez device, meaning they may not be optimal for the actual device noise profile [41].

6 Conclusion

This work benchmarked readout correction, parity post-selection, and zero-noise extrapolation, together with all their combinations, for VQE on the 1D transverse-field Ising model across 54 (N, h, L) configurations and two IBM noise models, with validation on real `ibm_fez` hardware. The central finding is that no universal optimal mitigation strategy exists: the choice of methods must be jointly informed by two axes — the noise composition of the quantum device and the symmetry structure of the target Hamiltonian’s ground state.

On the FakeBrisbane noise model, the optimal strategy was phase-dependent. In the ordered phase, readout + parity achieved a median relative error of $\tilde{\epsilon} \approx 0.03$ at $\sim 1.75\times$ overhead; in the disordered phase, parity failed and readout + ZNE became optimal at $\tilde{\epsilon} \approx 0.18$ and $\sim 3\times$ overhead; at criticality, all three methods contributed, reaching $\tilde{\epsilon} \approx 0.14$ at $\sim 5.8\times$ overhead. This absence of a universal

strategy implies that practical NISQ applications cannot rely on a fixed mitigation pipeline; the mitigation stack must instead be co-designed with both the target Hamiltonian and the hardware backend.

The effectiveness of ZNE specifically was found to be determined by the first axis: the device’s gate-to-readout noise ratio rather than by the algorithm or the physical system. The improvement fraction follows a sigmoidal dependence on the noise ratio r , with a crossover near $r \approx 0.4\text{--}0.5$. This was corroborated by the contrast between the Brisbane ($r = 0.31$) and Fez ($r = 0.51$) processors, across which ZNE shifted from ineffective to moderately effective. This relationship provides a predictive tool: a processor’s calibration data alone can indicate whether ZNE is worth deploying, before any benchmark is run. A complementary framework was developed independently by Sobrino et al. [12]; the two approaches agree that ZNE effectiveness is governed by the underlying noise structure rather than the algorithm.

Parity post-selection, by contrast, is governed by the second axis: the symmetry structure of the target ground state. It was highly effective in the ordered phase, where the optimiser was able to approximate the near-GHZ ground state, which sits almost entirely in the even-parity sector; but degraded toward ineffectiveness in the disordered phase, where the optimiser, limited by expressibility, was unable to approximate the ground state correctly and discard fractions approached 0.5. This is not a property of the device but of how cleanly the target state occupies a single symmetry sector and the expressibility of the ansatz used. Hence, parity post-selection is a natural complement to ZNE, which addresses the orthogonal, hardware determined axis.

Alternatively, readout correction was effective unconditionally, achieving a 10–15 percentage point error reduction across all phases at negligible overhead, and should be deployed as a default first step in any NISQ VQE implementation. The simulation methodology was validated against real `ibm_fez` hardware, where simulated and measured relative errors agreed within a few percentage points across all eight configurations, confirming FakeFez as a faithful noise model for the device.

Several limitations bound these conclusions. The COBYLA optimiser was run with only $n_{\text{restarts}} = 3$, which does not guarantee global optima and produces non-monotone N -dependence in some configurations, and $n_{\text{reps}} = 10$ evaluation repetitions leaves $\sim 30\%$ uncertainty on the SEM. The polynomial Richardson model assumes noise scales linearly under folding, an assumption only approximately satisfied for depolarising channels at high fold factors, which caps ZNE improvement at $\sim 40\%$ in simulation; on real hardware, T_1/T_2 decoherence imposes a further independent ceiling. Finally the hardware validation reuses simulator-optimised parameters and covers only a single (N, h, L) point.

These limitations motivate several extensions. Running the full benchmark on hardware, and across

a wider range of ansatz depths, would test the expressibility constraints identified here. Gate-folding could be replaced with pulse level ZNE [11], potentially avoiding the saturation observed in Section 4 and the decoherence ceiling on hardware. The per-qubit independence assumption could be relaxed to study correlated readout errors [42], and additional methods such as probabilistic error cancellation [9] would broaden the comparison. Most ambitiously, the predictive noise-ratio framework points toward adaptive mitigation: a scheme that reads a device’s calibration data and selects the optimal configuration automatically, without a full benchmark sweep — turning the central observation of this work into a practical tool.

Acknowledgements

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Appendix A Benchmark Implementation

The simulation code developed for this study follows the same pipeline for all (N, h, L) configurations. The workflow has four distinct stages. First, the TFIM Hamiltonian is constructed for the given (N, h) configuration using `SparsePauliOp` and diagonalised to obtain the E_0 reference energy. Then, $2N$ calibration circuits are executed on the noisy `AerSimulator` with $N_{\text{shots}}^{\text{cal}}$ to obtain the per-qubit assignment matrices A_q^{-1} , as described in Section 2.5.1. These matrices are computed once per (N, h, L) configuration and reused across all mitigation configurations containing readout correction. Following the calibration matrices, the hardware-efficient ansatz is constructed for (N, L) and optimised using COBYLA with n_{restarts} independent random initialisations. Each function evaluation uses the unmitigated cost function with $N_{\text{shots}}^{\text{opt}}$ shots. The restart yielding the lowest energy produces the optimised parameters θ^* . The optimisation is performed entirely under the noisy simulator, so the resulting θ^* reflects the noise-perturbed cost landscape rather than the ideal one [41]. Finally, the optimised parameters θ^* are evaluated across all eight mitigation configurations described in Section 2.6. For each configuration, n_{reps} independent repetitions are performed at $N_{\text{shots}}^{\text{eval}}$ shots with independent simulator seeds ensuring statistical independence. For ZNE-containing configurations, this evaluation is repeated at each noise scale factor $\lambda \in \{1, 3, 5\}$. The sample mean, SEM, relative error ϵ , COBYLA convergence history, optimal parameters θ^* , parity discard fraction f_{discard} , and effective circuit overhead Ω_{eff} (defined in Section 3.3) are recorded for each configuration.

The full benchmark sweeps 54 (N, h, L) combinations (6 system sizes $\times$ 3 field strengths $\times$ 3 ansatz depths). All random seeds are derived deterministically from a global base seed, making the entire benchmark fully reproducible.

Given that exact diagonalisation provides the theoretical ground state energy E_0 for any (N, h) configuration studied, it serves as an unambiguous reference to quantify the accuracy of VQE. Two error metrics are defined to study the performance of VQE and mitigation strategies: absolute error ΔE and relative error ϵ with respect to ED:

$$\Delta E = |E_{\text{VQE}} - E_{\text{ED}}|, \quad (\text{A.1})$$

and

$$\epsilon = \left| \frac{E_{\text{VQE}} - E_{\text{ED}}}{E_{\text{ED}}} \right|. \quad (\text{A.2})$$

The relative error ϵ is the primary error metric used in Section 3 [5]. It is dimensionless, scale independent, and allows meaningful comparison across different (N, h, L) configurations.

Throughout the study, each reported VQE energy is a statistical estimate. For a given set of optimised parameters θ^* , the energy is evaluated $n_{\text{reps}} = 10$ times. The simulator’s random seed is independently set, producing 10 statistically independent energy values $\{E_1, E_2, \dots, E_{10}\}$. The reported energy is the sample mean $\bar{E} = \frac{1}{n_{\text{reps}}} \sum_{i=1}^{n_{\text{reps}}} E_i$. The uncertainty is reported as the standard error of the mean:

$$\text{SEM} = \frac{s}{\sqrt{n_{\text{reps}}}}, \quad (\text{A.3})$$

where s is the sample standard deviation with Bessel’s correction. The SEM captures the effect of shot noise and stochastic gate noise. All error bars on figures in Section 3 correspond to ± 1 SEM. With $n_{\text{reps}} = 10$, the SEM itself has a relative uncertainty of $\sim 30\%$. Reducing this to 10% would require $n_{\text{reps}} \sim 100$. Higher precision can be achieved by increasing n_{reps} , at the expense of increasing computational cost.

In addition to the relative error ϵ for individual configurations, the effectiveness of each mitigation strategy is quantified by the error reduction, defined as $\Delta\epsilon = \epsilon_{\text{raw}} - \epsilon_{\text{mitigated}}$. A positive $\Delta\epsilon$ indicates that mitigation has improved accuracy. This metric isolates the contribution of error mitigation from baseline VQE performance and is also used in Section 3 to compare mitigation strategies across (N, h, L) configurations.

Parameter	Symbol	Value
System sizes	N	$\{2, 4, 6, 8, 10, 12\}$
Field strengths	h	$\{0.5, 1.0, 2.0\}$
Coupling constant	J	1.0
Ansatz depths	L	$\{1, 2, 3\}$
Evaluation shots	$N_{\text{shots}}^{\text{eval}}$	8192
Optimisation shots	$N_{\text{shots}}^{\text{opt}}$	1024
Calibration shots	$N_{\text{shots}}^{\text{cal}}$	2048
Repetitions	n_{reps}	10
COBYLA restarts	n_{restarts}	3
Max iterations	maxiter	600
Trust region radius	ρ_0	$\pi/4$
ZNE scale factors	λ	$\{1, 3, 5\}$
Global seed	–	42

Table A.1: Summary of benchmark hyperparameters. Individual parameters are introduced and justified in Sections 2.3.3, 2.5.1 and 2.5.3 and Appendix A.

Appendix B VQE Convergence and Optimiser Behaviour

The convergence plot in Figure B.1 shows the relative error of the circuit evaluation, $|E_{\text{iter}} - E_0|/|E_0|$ as a function of COBYLA iteration for the best optimisation restart on the FakeBrisbane backend. All convergence curves for $N \leq 10$, $L \leq 2$ plateau before the $\text{maxiter} = 600$ limit, confirming convergence. At the extremal configuration in Figure B.1i $N = 12$, $L = 3$, the curves are still slowly descending at ~ 450 iterations, which justifies the choice of $\text{maxiter} = 600$.

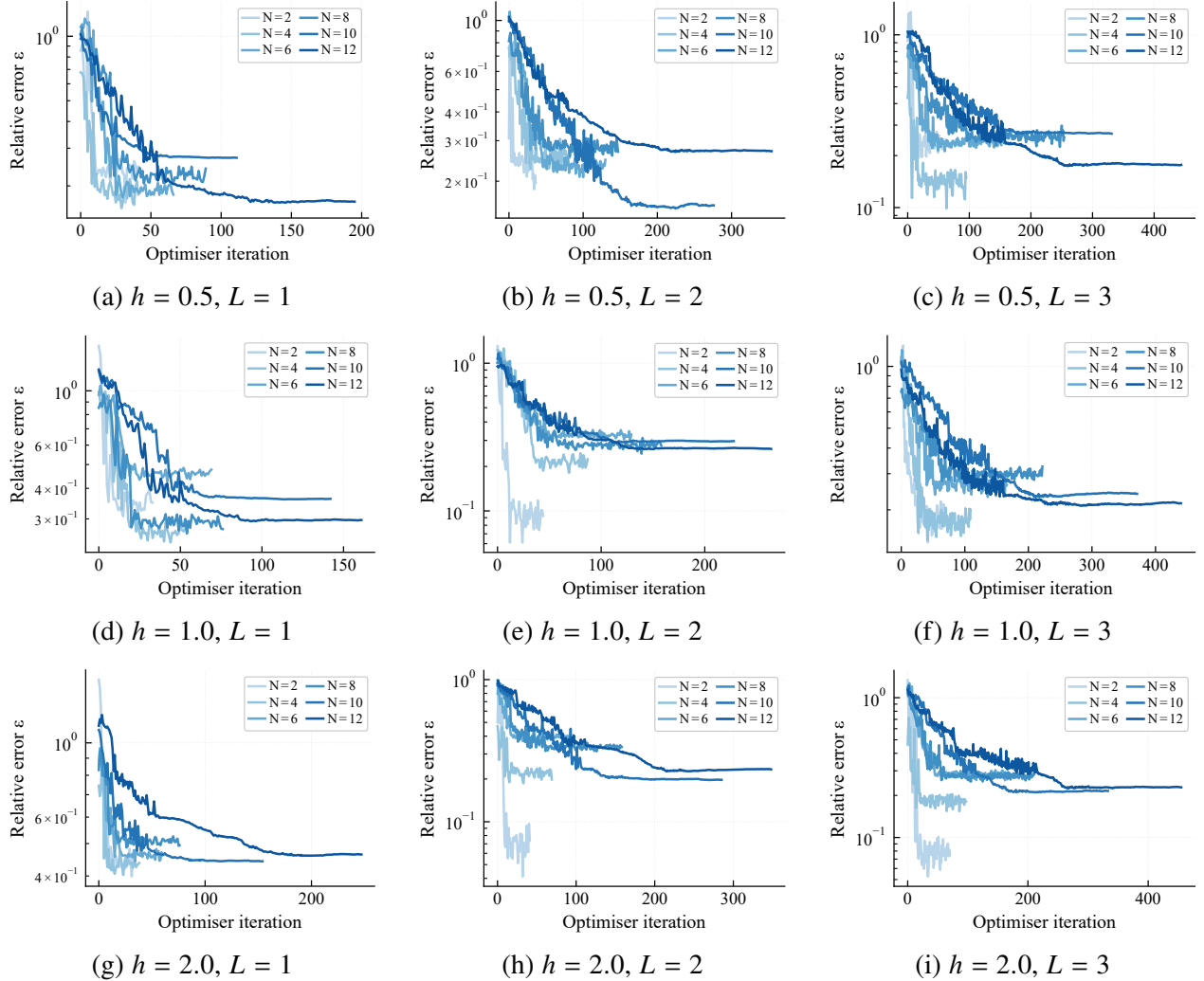


Figure B.1: Relative error ϵ during COBYLA optimisation as a function of iteration number, for the best restart at each (N, h, L) combination on FakeBrisbane. Rows correspond to transverse field strength $h \in \{0.5, 1.0, 2.0\}$ (top to bottom) and columns to ansatz depth $L \in \{1, 2, 3\}$ (left to right). Colour intensity increases with system size N from light ($N = 2$) to dark ($N = 12$). All curves for $N \leq 10$ or $L \leq 2$ plateau well before $\text{maxiter} = 600$; the extremal configuration ($N = 12$, $L = 3$, panel i) is still slowly descending at iteration ~ 450 , justifying the chosen limit.

The convergence profiles are qualitatively similar across all three quantum phases, despite the

theoretically harder landscape at criticality. This is attributed to shot noise from $N_{\text{shots}}^{\text{opt}} = 1024$ dominating the landscape seen by COBYLA, washing out the underlying phase-dependent difference in the cost function. This is consistent with the stochastic nature of the noisy objective function discussed in Section 2.3.3.

Additionally, the relative error does not always increase monotonically with N . At certain (N, h, L) configurations, particularly in the ordered phase, the error drops unexpectedly. This also reflects the stochastic nature of COBYLA with $n_{\text{restarts}} = 3$, as the optimiser can occasionally find near-global optima. This is especially consistent with the ordered ground state, whose GHZ-like structure has both a deep global minimum and shallow local minima. This is not a simulation artifact, but a feature of derivative-free optimisation on noisy landscapes with limited restarts.

Figure B.1 also shows that all energies remain above the exact value, such that $E_{\text{VQE}} \geq E_0$ is empirically observed, preserving the variational upper bound. As discussed in Section 2.4, this is consistent with the dominant noise sources (readout and depolarising errors) both biasing the energy upward. Despite this, brief dips below E_0 in individual function evaluations are possible, as shot noise can momentarily push a single evaluation below the true ground state energy. These do not constitute violations of the variational principle, which applies to expectation values, not single-shot estimates.

Appendix C Code Availability

The complete source code for this work is available at <https://github.com/MartiNadal/VQE-error-mitigation-qiskit>. All code was developed independently by the author. The implementations of readout error correction, parity post-selection, and zero-noise extrapolation follow the methods described in Section 2.5, which are based on the references cited therein. No code was adapted from external repositories or third-party implementations.

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